

Outline for section 1

Why quantum computing?

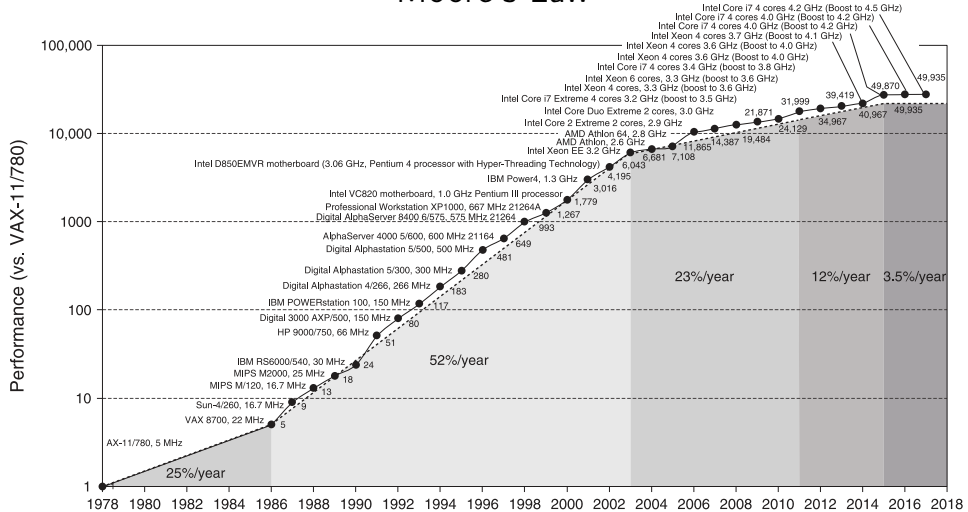
- Moore's law

- Amdahl's law

- Patents in quantum computing and quantum machine learning

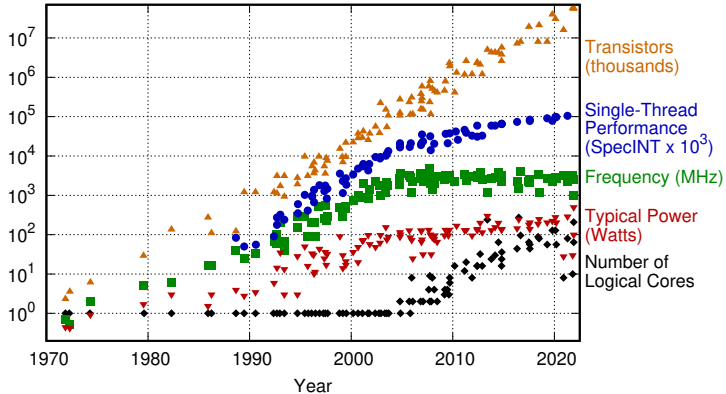
Our World
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Moore's Law



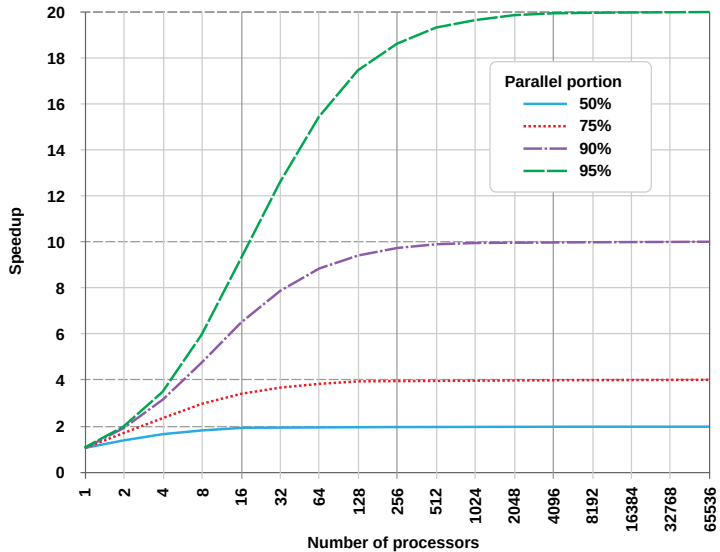
Source: Hennessy, John L., and David A. Patterson. Computer architecture: a quantitative approach. Elsevier, 2011. 6th edition.

50 Years of Microprocessor Trend Data



Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2021 by K. Rupp

Amdahl's Law



Quantum computing

Insight report

January 2023

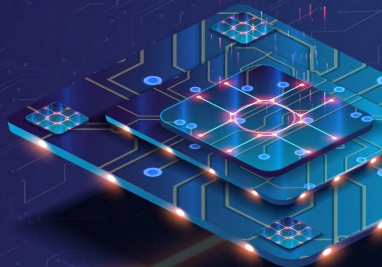
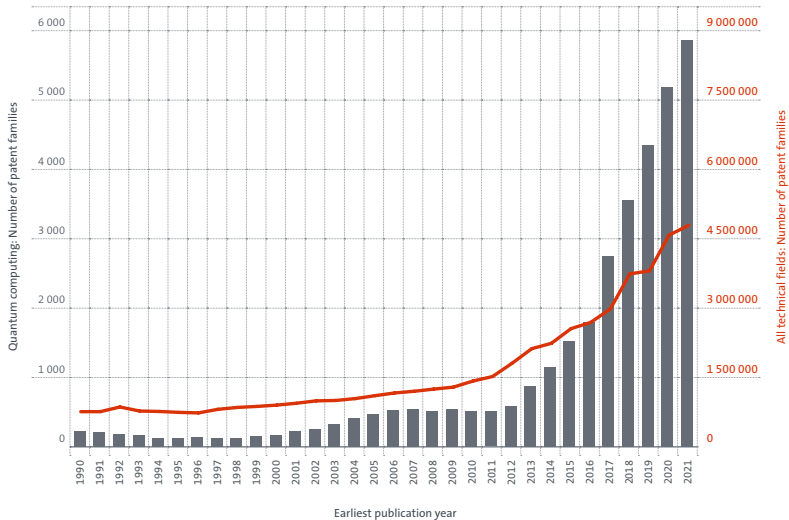


Figure 2

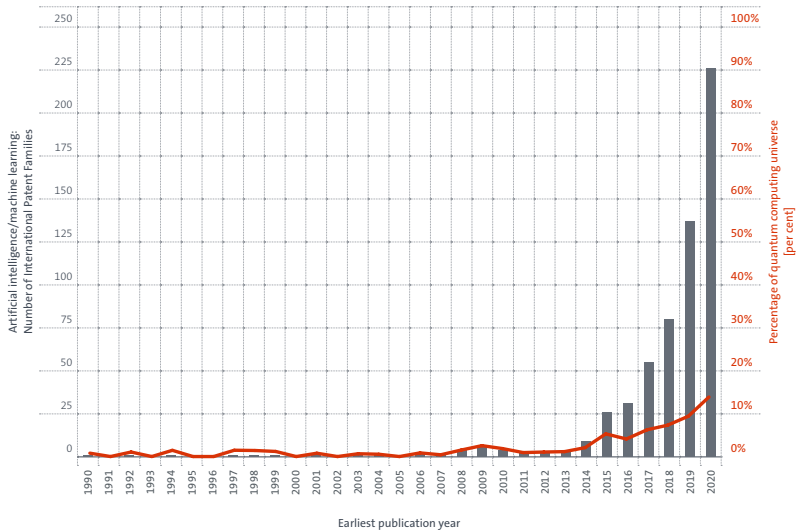
Number of DOCDB patent families per earliest publication year in the field of quantum computing



Source: authors' calculations

Figure 16

Number of inventions per earliest publication year related to quantum computing and artificial intelligence/machine learning



Source: authors' calculations

Outline for section 2

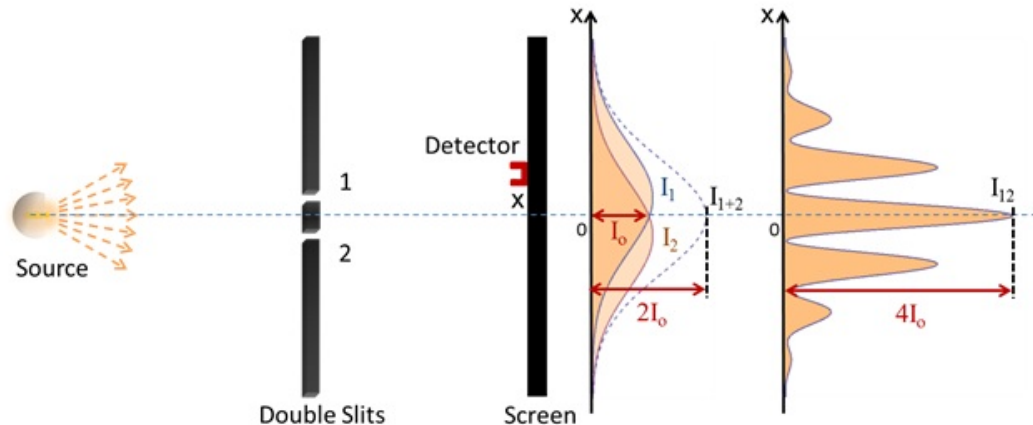
Introduction to the quantum world

- Interference

- Mach-Zender interferometer

- Elitzur-Vaidman bomb testing problem

Interference

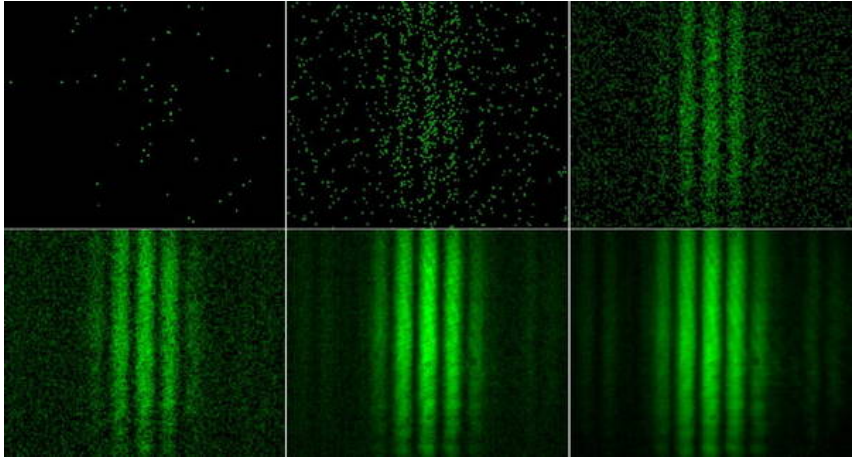


Classical: $I_{1+2} = I_1 + I_2$

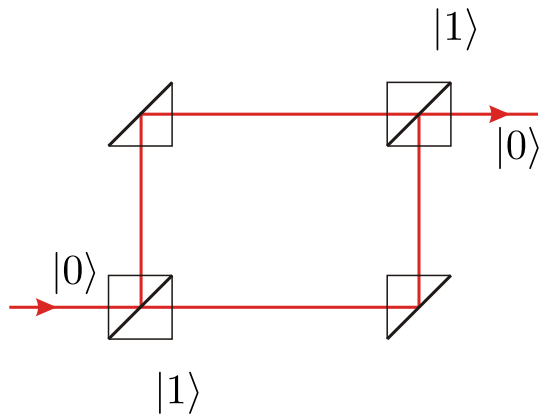
Quantum: $I_{12} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\theta_1 - \theta_2)$

$I_{12} = |\alpha_1 + \alpha_2|^2, \alpha_i \in \mathbb{C}, \alpha_i = |\alpha_i| e^{i\theta_i}.$

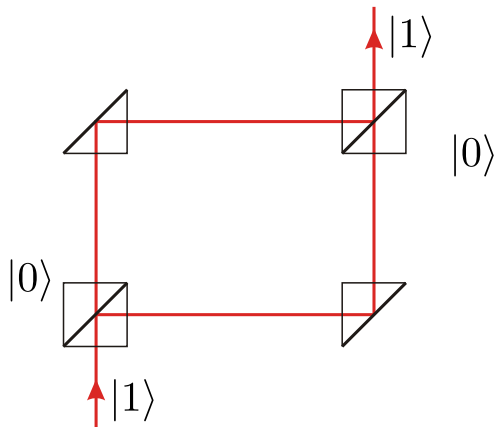
Interference



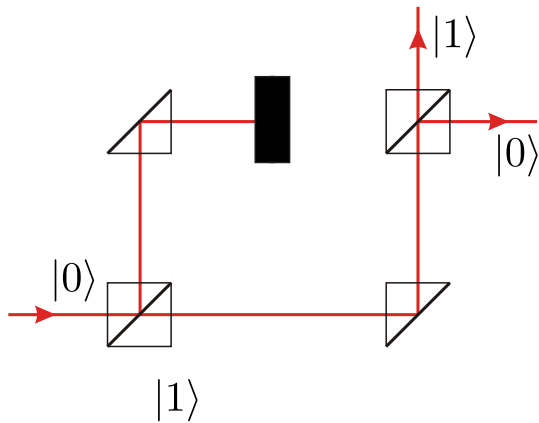
Mach-Zender interferometer I



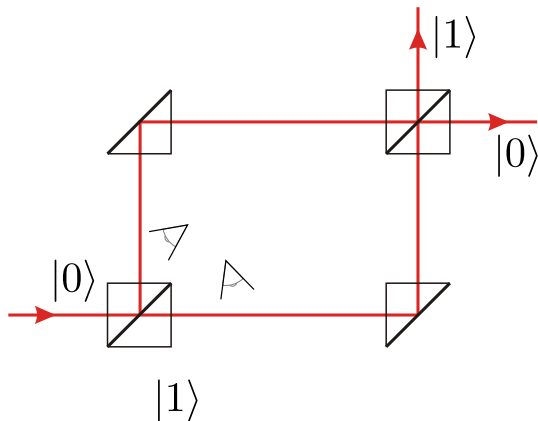
Mach-Zender interferometer II



Mach-Zender interferometer III



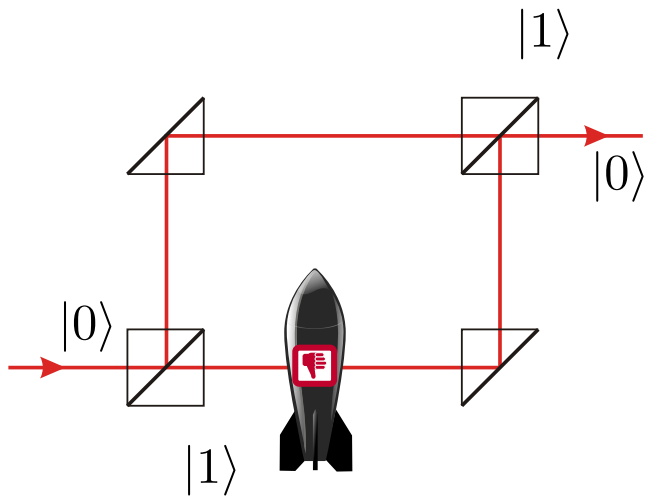
Mach-Zender interferometer IV



Elitzur-Vaidman bomb testing problem I

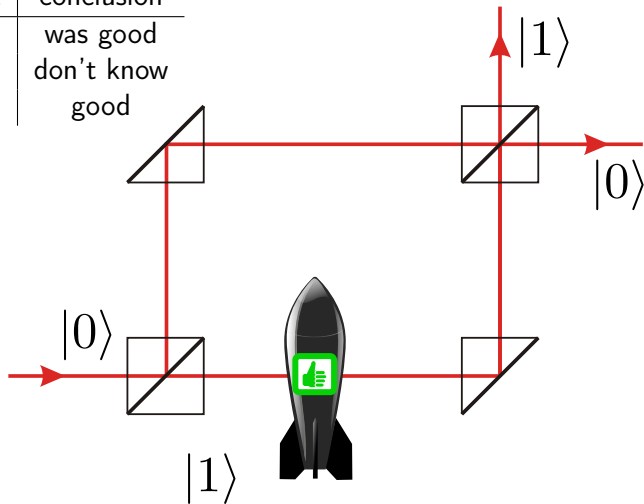


Elitzur-Vaidman bomb testing problem II



Elitzur-Vaidman bomb testing problem III

p	explosion	result	conclusion
1/2	yes	—	was good
1/4	no	0	don't know
1/4	no	1	good



Outline for section 3

Bibliography

Bibliography I

-  Gawron, Piotr, Michał Cholewa, and Katarzyna Kara. *Rewolucja stanu–fantastyczne wprowadzenie do informatyki kwantowej*. Instytut Informatyki Teoretycznej i Stosowanej Polskiej Akademii Nauk, 2016.
-  McMahon, David. *Quantum Computing Explained*. John Wiley & Sons, Inc., Nov. 2007.
-  Nielsen, Michael A and Isaac Chuang. *Quantum computation and quantum information*. 2002.
-  Schuld, Maria and Francesco Petruccione. *Machine Learning with Quantum Computers*. Springer International Publishing, 2021.

Outline for section 5

Building blocks of quantum world

- Classical and quantum computing

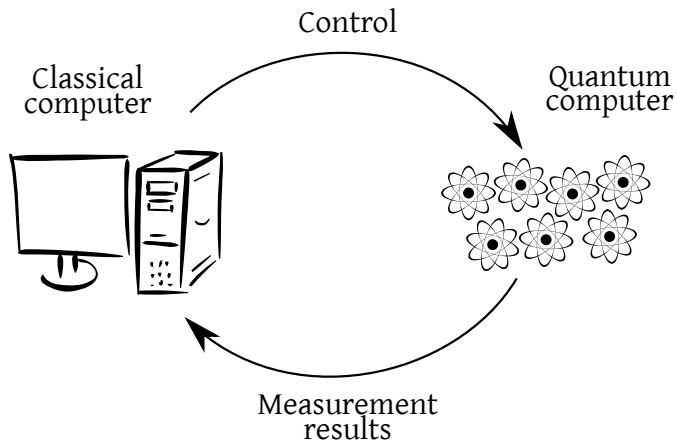
- Quantum states

- Quantum measurements

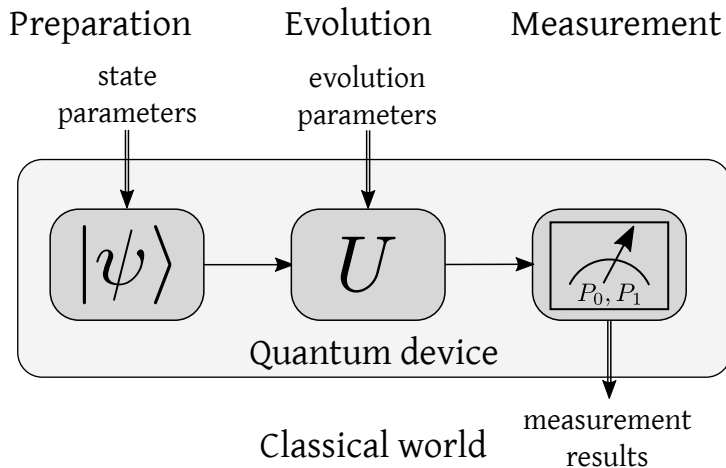
- Quantum channels

- Entanglement and Bell states

Quantum computers are not self-sufficient



Scheme of a quantum device



Bits and qubits

Bit

0 or 1

Qubit

$$\alpha |0\rangle + \beta |1\rangle$$

Mathematical properties of qubits

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$|\alpha|^2 + |\beta|^2 = 1$$

$$e^{i\phi} |\psi\rangle \equiv |\psi\rangle$$

Dirac bra-ket notation

Let

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad |\phi\rangle = \begin{bmatrix} \gamma \\ \delta \end{bmatrix}, \quad \langle\psi| = |\psi\rangle^\dagger = [\bar{\alpha}, \bar{\beta}].$$

Properties:

► inner (scalar) product

$$\langle\psi|\phi\rangle = [\bar{\alpha}, \bar{\beta}] \begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \bar{\alpha}\gamma + \bar{\beta}\delta,$$

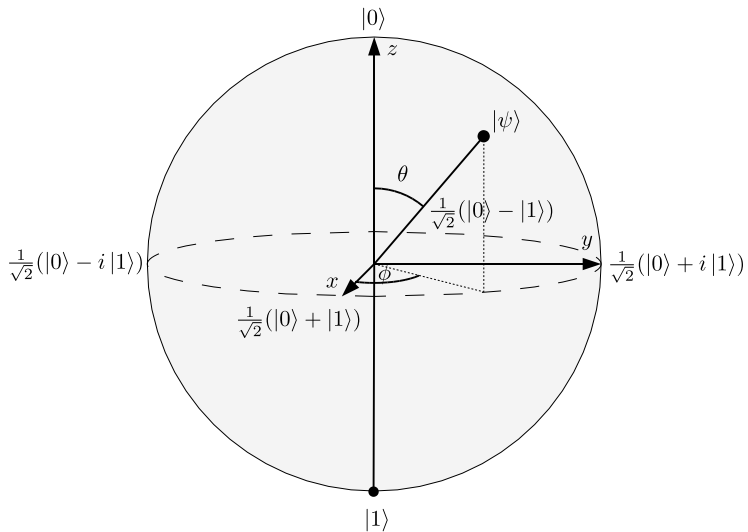
► length

$$\sqrt{\langle\psi|\psi\rangle} = \sqrt{|\alpha|^2 + |\beta|^2} = 1,$$

► outer product

$$|\psi\rangle\langle\phi| = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} [\bar{\gamma} \quad \bar{\delta}] = \begin{bmatrix} \alpha\bar{\gamma} & \alpha\bar{\delta} \\ \beta\bar{\gamma} & \beta\bar{\delta} \end{bmatrix}.$$

Representation of qubits



Measurement

Quantum measurement is a function μ :

- ▶ from finite set of measurements outcomes $A = \{a_i\}_{i=0}^{n-1}$
- ▶ to set the of projection operators $P = \{P_i\}_{i=0}^{n-1}$ (where projections satisfy $P_i^2 = P_i$) such that

$$\sum_{i=0}^{n-1} P_i = \mathbb{I}.$$

Measurement

The **probability of measuring outcome** a_i given the quantum system in state $|\psi\rangle$ is

$$p(a_i) = \langle \psi | P_i | \psi \rangle .$$

The state of the quantum system **after the measurement** outcome a_i was obtained becomes

$$|\psi\rangle_{a_i} = \frac{P_i |\psi\rangle}{\sqrt{\langle \psi | P_i | \psi \rangle}} .$$

Example 1 — part I

We will measure the state

$$|\psi\rangle = \sqrt{0.7} |0\rangle + \sqrt{0.3i} |1\rangle$$

by a measurement

$$A = \{0, 1\}, \quad P = \left\{ P_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, P_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

Probabilities of obtaining outcomes 0 and 1

$$p(0) = \langle \psi | P_0 | \psi \rangle = \begin{bmatrix} \sqrt{0.7} & -\sqrt{0.3i} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{0.7} \\ \sqrt{0.3i} \end{bmatrix} = 0.7,$$

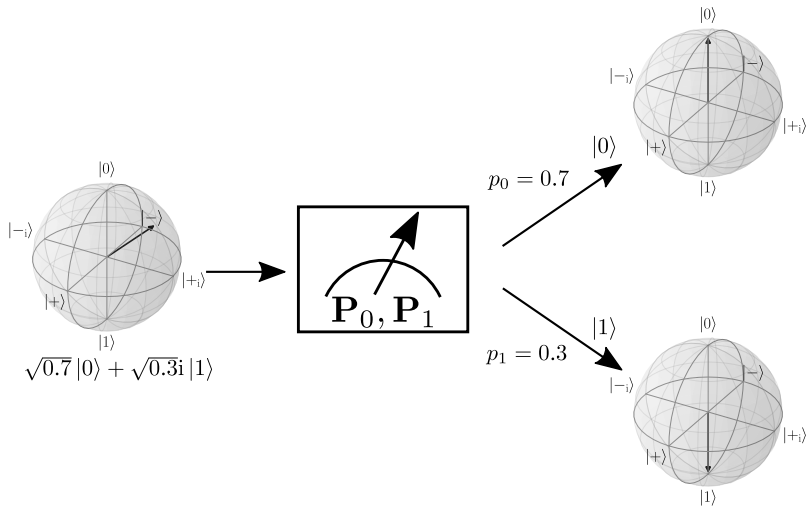
$$p(1) = \langle \psi | P_1 | \psi \rangle = \begin{bmatrix} \sqrt{0.7} & -\sqrt{0.3i} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{0.7} \\ \sqrt{0.3i} \end{bmatrix} = 0.3.$$

Example 1 — part II

Conditional states after the measurement

$$\begin{aligned} |\psi\rangle_0 &= \frac{P_0 |\psi\rangle}{\sqrt{\langle\psi| P_0 |\psi\rangle}} = \frac{1}{\sqrt{0.7}} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{0.7} \\ \sqrt{0.3}i \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ |\psi\rangle_1 &= \frac{P_1 |\psi\rangle}{\sqrt{\langle\psi| P_1 |\psi\rangle}} = \frac{1}{\sqrt{0.3}} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{0.7} \\ \sqrt{0.3}i \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}$$

Example 1 — part III



Example 2 — part I

We will measure the state

$$|\psi\rangle = \sqrt{0.7}|0\rangle + \sqrt{0.3}\mathrm{i}|1\rangle$$

by a measurement

$$A = \{-\mathrm{i}, +\mathrm{i}\}, \quad Q = \left\{ Q_{-\mathrm{i}} = \frac{1}{2} \begin{bmatrix} 1 & \mathrm{i} \\ -\mathrm{i} & 1 \end{bmatrix}, \quad Q_{+\mathrm{i}} = \frac{1}{2} \begin{bmatrix} 1 & -\mathrm{i} \\ \mathrm{i} & 1 \end{bmatrix} \right\}.$$

Probabilities of obtaining outcomes $-\mathrm{i}$ and $+\mathrm{i}$

$$p(-\mathrm{i}) = \langle\psi| Q_{-\mathrm{i}} |\psi\rangle = [\sqrt{0.7} \quad -\sqrt{0.3}\mathrm{i}] \frac{1}{2} \begin{bmatrix} 1 & \mathrm{i} \\ -\mathrm{i} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{0.7} \\ \sqrt{0.3}\mathrm{i} \end{bmatrix} = \frac{1}{10}(5 - \sqrt{21}) \approx 0.042,$$

$$p(+\mathrm{i}) = \langle\psi| Q_{+\mathrm{i}} |\psi\rangle = [\sqrt{0.7} \quad -\sqrt{0.3}\mathrm{i}] \frac{1}{2} \begin{bmatrix} 1 & -\mathrm{i} \\ \mathrm{i} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{0.7} \\ \sqrt{0.3}\mathrm{i} \end{bmatrix} = \frac{1}{10}(5 + \sqrt{21}) \approx 0.958.$$

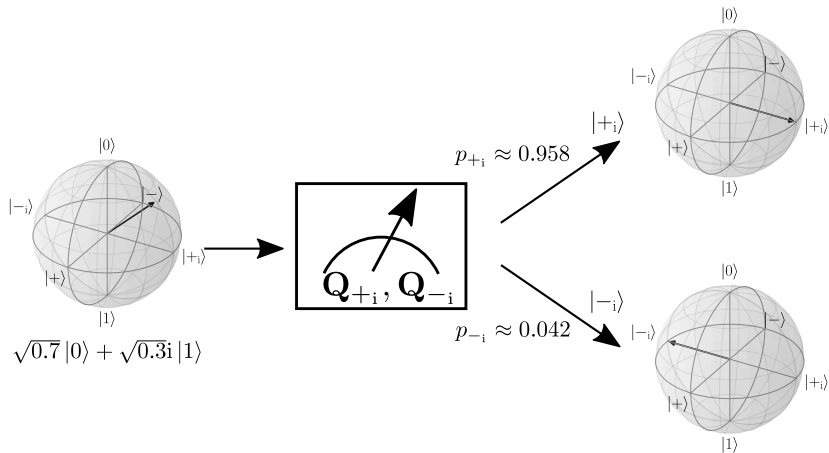
Example 2 — part II

Conditional states after the measurement

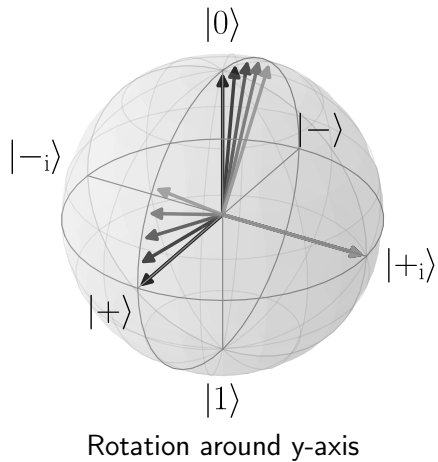
$$|\psi\rangle_{-i} = \frac{Q_{-i} |\psi\rangle}{\sqrt{\langle\psi| Q_{-i} |\psi\rangle}} = \frac{1}{\sqrt{\frac{1}{10}(5 - \sqrt{21})}} \frac{1}{2} \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \begin{bmatrix} \sqrt{0.7} \\ \sqrt{0.3} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} = |-i\rangle$$

$$|\psi\rangle_{+i} = \frac{Q_{+i} |\psi\rangle}{\sqrt{\langle\psi| Q_{+i} |\psi\rangle}} = \frac{1}{\sqrt{\frac{1}{10}(5 + \sqrt{21})}} \frac{1}{2} \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} \begin{bmatrix} \sqrt{0.7} \\ \sqrt{0.3} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} = |+i\rangle$$

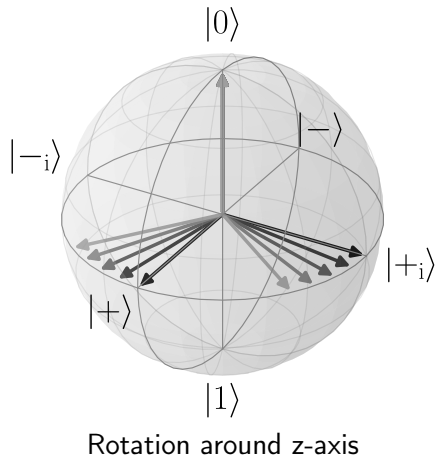
Example 2 — part III



Quantum evolution



Quantum evolution



Composition of quantum systems

Tensor product for two qubit states

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha |0\rangle + \beta |1\rangle ,$$

$$|\phi\rangle = \begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \gamma |0\rangle + \delta |1\rangle ,$$

and their joint state in $\mathbb{C}^2 \otimes \mathbb{C}^2$

$$|\psi\rangle \otimes |\phi\rangle = \begin{bmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{bmatrix} ,$$

$$|\psi\rangle \otimes |\phi\rangle = \alpha\gamma |0\rangle \otimes |0\rangle + \alpha\delta |0\rangle \otimes |1\rangle + \beta\gamma |1\rangle \otimes |0\rangle + \beta\delta |1\rangle \otimes |1\rangle ,$$

$$|\psi\phi\rangle = \alpha\gamma |00\rangle + \alpha\delta |01\rangle + \beta\gamma |10\rangle + \beta\delta |11\rangle .$$

Entanglement

Recall the product state

$$|\psi\phi\rangle = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle,$$

and consider another state

$$|\Psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle.$$

Question

Can we always find coefficients which satisfy

$$a = \alpha\gamma$$

$$b = \alpha\delta$$

$$c = \beta\gamma$$

$$d = \beta\delta?$$

Entanglement

Bell State

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

$$|\Phi^+\rangle \stackrel{?}{=} \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle,$$

Contradiction

$$\alpha\gamma = \beta\delta = \frac{1}{\sqrt{2}},$$

$$\alpha\delta = \beta\gamma = 0.$$

Outline for section 6

Gate model of computation

- Classical gates

- Single qubit gates

- 2-qubit gates

- Deutsch algorithm

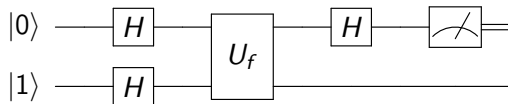
- Teleportation

Gate model

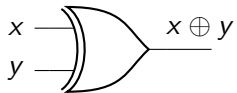
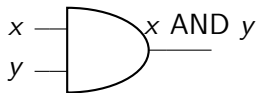
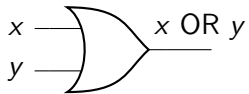
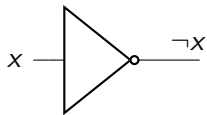
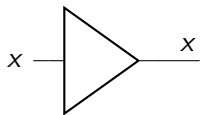
How to perform computations?

- ▶ Prepare the system in a known state, eg.: $|0\rangle \otimes |1\rangle$.
- ▶ Apply quantum gates and partial measurements on the state.
- ▶ Perform the measurement.
- ▶ Analyze the result and post-processing.

Example of a quantum circuit



Classical gates



Single qubit gate - unitary gate

How to denote a quantum single-qubit gate?

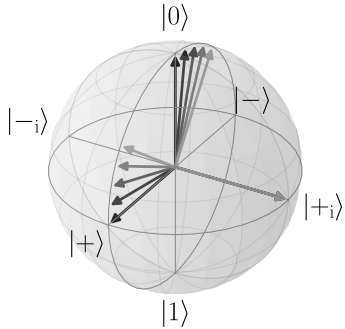
$$|\psi\rangle \xrightarrow{U} U|\psi\rangle$$

Quantum evolution is time-reversible!

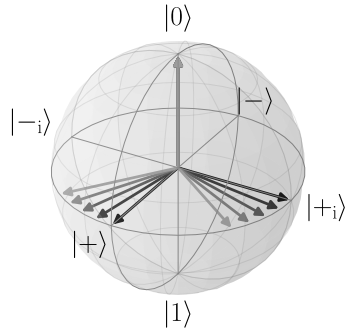
$$U|\psi_{t=0}\rangle = |\psi_{t=1}\rangle$$

$$U^\dagger |\psi_{t=1}\rangle = |\psi_{t=0}\rangle$$

Rotations



$$R_y(\gamma) = \begin{bmatrix} \cos\left(\frac{\gamma}{2}\right) & \sin\left(\frac{\gamma}{2}\right) \\ -\sin\left(\frac{\gamma}{2}\right) & \cos\left(\frac{\gamma}{2}\right) \end{bmatrix}$$



$$R_z(\beta) = \begin{bmatrix} e^{i\frac{\beta}{2}} & 0 \\ 0 & -e^{i\frac{\beta}{2}} \end{bmatrix}$$

Important examples of single-qubit gates

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{I} \longrightarrow \alpha|0\rangle + \beta|1\rangle \qquad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{X} \longrightarrow \beta|0\rangle + \alpha|1\rangle \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

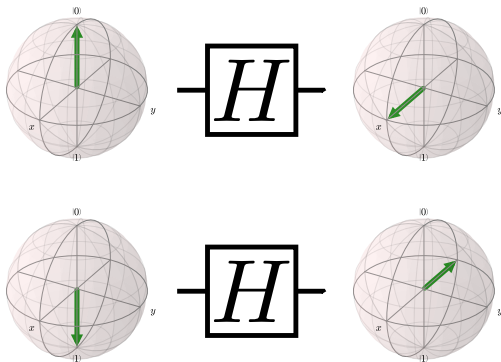
$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{Y} \longrightarrow -\beta i|0\rangle + \alpha i|1\rangle \qquad \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{Z} \longrightarrow \alpha|0\rangle - \beta|1\rangle \qquad \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{P(\phi)} \longrightarrow \alpha|0\rangle + e^{i\phi}\beta|1\rangle \qquad \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix}$$

Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$



Decomposition of a unitary gate

Definition

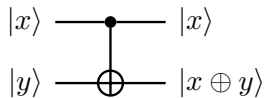
A square matrix is called *unitary* iff

$$UU^\dagger = U^\dagger U = \mathbb{1}.$$

Every unitary gate U can be written as

$$U = R_Z(\psi)R_Y(\theta)R_Z(\Delta) = e^{i\varphi/2} \begin{bmatrix} e^{i\psi} & 0 \\ 0 & e^{-i\psi} \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} e^{i\Delta} & 0 \\ 0 & e^{-i\Delta} \end{bmatrix}.$$

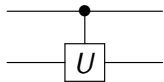
CNOT gate



in ₁	in ₂	out ₁	out ₂
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} : \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} \rightarrow \begin{bmatrix} \alpha \\ \beta \\ \delta \\ \gamma \end{bmatrix}$$

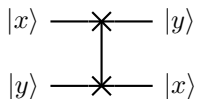
Controlled unitary



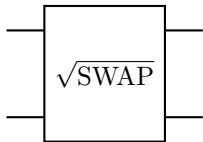
$$U = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix}$$

$$CU_1^2 = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & u_{00} & u_{01} \\ 0 & 0 & u_{10} & u_{11} \end{bmatrix}$$

Swap gate

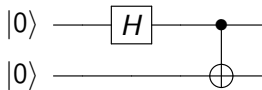


$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} : \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} \rightarrow \begin{bmatrix} \alpha \\ \gamma \\ \beta \\ \delta \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1+i}{2} & \frac{1-i}{2} & 0 \\ 0 & \frac{1-i}{2} & \frac{1+i}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} : \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} \rightarrow \begin{bmatrix} \alpha \\ \frac{1+i}{2}\beta + \frac{1-i}{2}\gamma \\ \frac{1-i}{2}\beta + \frac{1+i}{2}\gamma \\ \delta \end{bmatrix}$$

Example of a circuit



$$1. (H \otimes I) |0\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$2. \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Deutsch algorithm – problem formulation

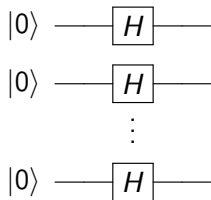
- ▶ Consider a function $f : \{0, 1\}^N \rightarrow \{0, 1\}$ that is either constant or balanced (i.e. half of its values are 0s and the other half are 1s).
- ▶ For $N = 1$ each function is either constant or balanced.

	$f(0)$	$f(1)$	
f_0	0	0	constant
f_1	1	1	
identity	0	1	balanced
bit flip	1	0	

Problem

Given f decide if it is constant or balanced.

Hadamard transform



Hadamard transform creates an equal superposition of all possible states of the system.

For two qubits we have:

$$\begin{aligned}(H \otimes H) |00\rangle &= \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \\ &= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle).\end{aligned}$$

And more generally:

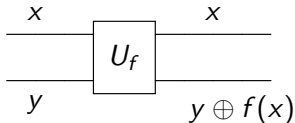
$$H^{\otimes N} |0\rangle^{\otimes N} = \frac{1}{\sqrt{2^N}} \sum_{x \in \{0,1\}^N} |x\rangle.$$

Black box

To implement Deutsch's algorithm we need a black box computing the following:

$$U_f |x, y\rangle = |x, y \oplus f(x)\rangle$$

x	y	$y \oplus x$
0	0	0
0	1	1
1	0	1
1	1	0



Black box

We can express U_f in a way that will be useful later. For instance

$$U_f |00\rangle = |0, 0 \oplus f(0)\rangle = \begin{cases} |00\rangle & \text{if } f(0) = 0 \\ |01\rangle & \text{if } f(0) = 1, \end{cases}$$

and hence

$$U_f |00\rangle = (1 - f(0)) |00\rangle + f(0) |01\rangle .$$

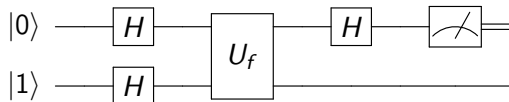
Also

$$U_f |01\rangle = f(0) |00\rangle + (1 - f(0)) |01\rangle ,$$

$$U_f |10\rangle = (1 - f(1)) |10\rangle + f(1) |11\rangle ,$$

$$U_f |11\rangle = f(1) |10\rangle + (1 - f(1)) |11\rangle .$$

Deutsch algorithm



Algorithm

1. Apply the Hadamard transform to the input state $|01\rangle$.
2. Apply U_f to the product state.
3. Apply the Hadamard gate to the first qubit.
4. Measure the first qubit.

Deutsch algorithm — part I

- Apply the Hadamard transform to the input state $|01\rangle$.

$$\begin{aligned} |\psi_1\rangle &= (H \otimes H) |01\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle) \end{aligned}$$

- Apply U_f to the product state.

$$\begin{aligned} |\psi_2\rangle &= U_f |\psi_1\rangle = \frac{1}{2} (U_f |00\rangle - U_f |01\rangle + U_f |10\rangle - U_f |11\rangle) \\ &= \frac{1 - 2f(0)}{2} |0\rangle \otimes (|0\rangle - |1\rangle) + \frac{1 - 2f(1)}{2} |1\rangle \otimes (|0\rangle - |1\rangle) \end{aligned}$$

Deutsch algorithm — part II

- ▶ Apply the Hadamard gate to the first qubit.

$$\begin{aligned} |\psi_3\rangle &= (H \otimes I) |\psi_2\rangle = (1 - f(0) - f(1)) |0\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ &\quad + (f(1) - f(0)) |1\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \end{aligned}$$

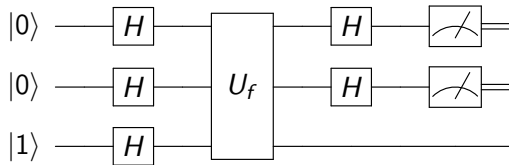
- ▶ Measure the first qubit.

Deutsch-Jozsa algorithm

Question

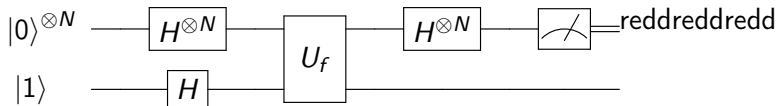
Is binary function $f : \{0, 1\}^N \rightarrow \{0, 1\}$ constant or balanced?

Circuit for $N = 2$



where $U_f |x, x, y\rangle = |x, x, y \oplus f(x)\rangle$.

Deutsch-Jozsa algorithm



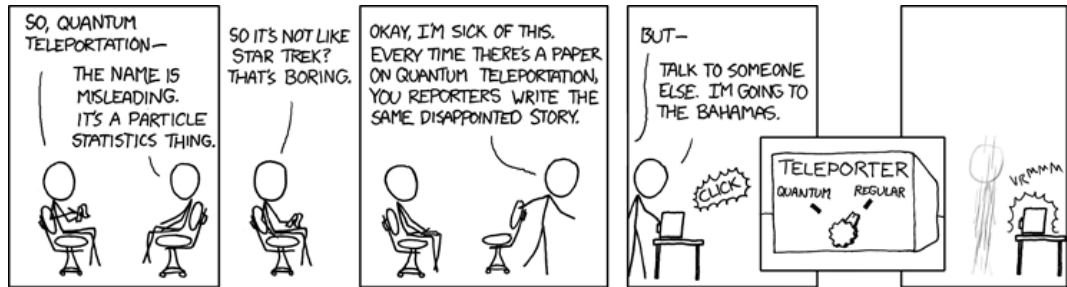
Algorithm

1. Apply the Hadamard transform to the input state $|0\rangle \otimes \dots \otimes |0\rangle \otimes |1\rangle$.
2. Apply U_f to the product state, where

$$U_f |x, \dots, x, y\rangle = |x, \dots, x, y \oplus f(x)\rangle.$$

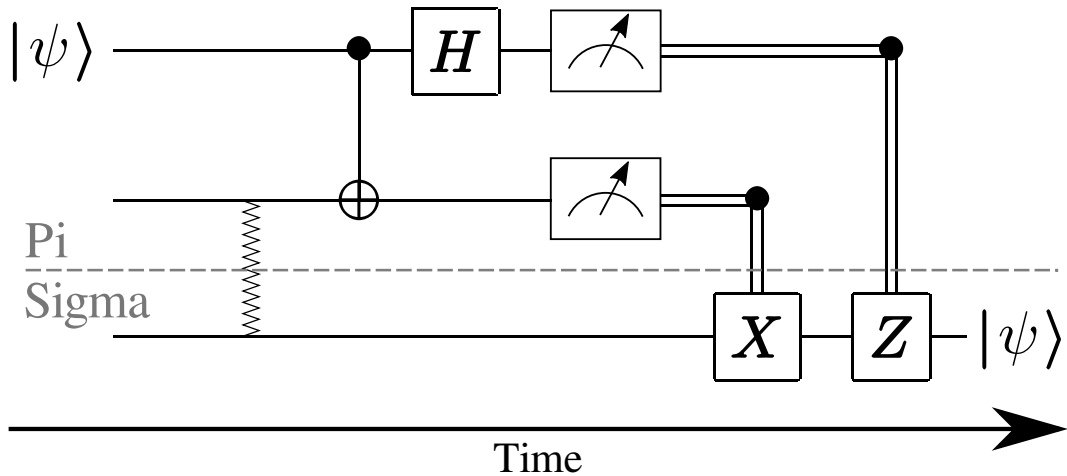
3. Apply the Hadamard gate to the first N qubits.
4. Measure the first N qubits.

Teleportation — vision

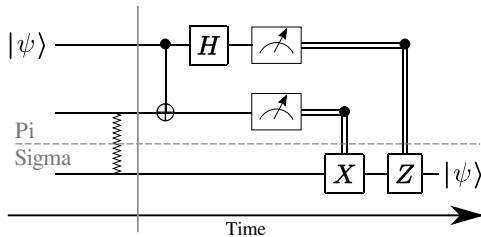


Source: <http://xkcd.com/465/> ©Randall Munroe 2015, CC-BY/2.5.

Teleportation — theory

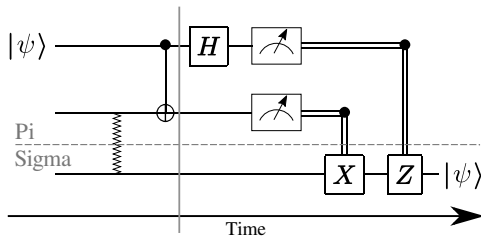


Teleportation — theory



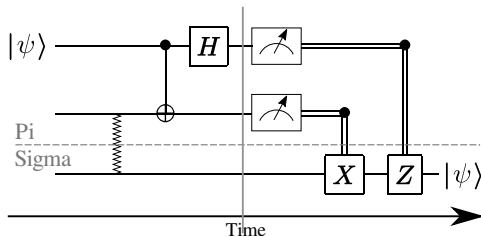
$$\begin{aligned}
 |\psi_{t=0}\rangle &= |\psi\rangle \otimes |\Phi^+\rangle = \\
 &= (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle) = \\
 &= \frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle)
 \end{aligned}$$

Teleportation — theory



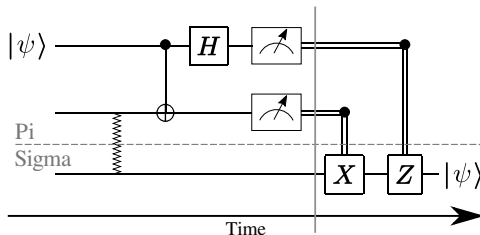
$$\begin{aligned}
 |\psi_{t=1}\rangle &= (\text{CNOT}_1^2 \otimes \mathbb{I}) \frac{1}{\sqrt{2}} (\alpha |000\rangle + \alpha |011\rangle + \beta |100\rangle + \beta |111\rangle) = \\
 &= \frac{1}{\sqrt{2}} (\alpha |000\rangle + \alpha |011\rangle + \beta |110\rangle + \beta |101\rangle).
 \end{aligned}$$

Teleportation — theory



$$\begin{aligned}
 |\psi_{t=2}\rangle &= (H \otimes I \otimes I) \frac{1}{\sqrt{2}} (\alpha |000\rangle + \alpha |011\rangle + \beta |110\rangle + \beta |101\rangle) = \\
 &= \frac{1}{\sqrt{2}} \left(\alpha \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes (|00\rangle + |11\rangle) + \beta \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \otimes (|10\rangle + |01\rangle) \right) = \\
 &= \frac{1}{2} \alpha (|000\rangle + |100\rangle + |011\rangle + |111\rangle) + \frac{1}{2} \beta (|010\rangle + |001\rangle - |110\rangle - |101\rangle)
 \end{aligned}$$

Teleportation — theory



$$\mathcal{P} = \{P_{00?} = |0\rangle\langle 0| \otimes |0\rangle\langle 0| \otimes \mathbb{I}, P_{01?} = |0\rangle\langle 0| \otimes |1\rangle\langle 1| \otimes \mathbb{I}, \\ P_{10?} = |1\rangle\langle 1| \otimes |0\rangle\langle 0| \otimes \mathbb{I}, P_{11?} = |1\rangle\langle 1| \otimes |1\rangle\langle 1| \otimes \mathbb{I}\}$$

- ▶ result 00?: $\frac{1}{\sqrt{2}} (\alpha |000\rangle + \beta |001\rangle) = \frac{1}{\sqrt{2}} (|00\rangle \otimes |\psi\rangle),$
- ▶ result 01?: $\frac{1}{\sqrt{2}} (\alpha |011\rangle + \beta |010\rangle) = \frac{1}{\sqrt{2}} (|01\rangle \otimes Z |\psi\rangle),$
- ▶ result 10?: $\frac{1}{\sqrt{2}} (\alpha |100\rangle - \beta |101\rangle) = \frac{1}{\sqrt{2}} (|10\rangle \otimes X |\psi\rangle),$
- ▶ result 11?: $\frac{1}{\sqrt{2}} (\alpha |111\rangle - \beta |110\rangle) = \frac{1}{\sqrt{2}} (|11\rangle \otimes XZ |\psi\rangle)$

Outline for section 7

From algorithm to physical realization

- Programming stack

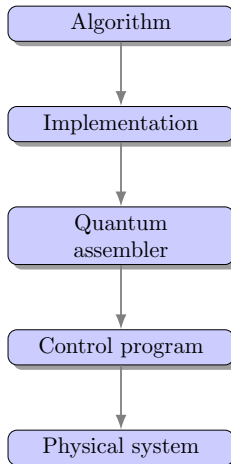
- QuTiP implementation

- Quantum assembler

- Quantum physical system

- Steering the program

Programming stack



Implementation

QuTiP implementation of Deutsch's algorithm for function $f(0) = 1, f(1) = 0$.

```
qc = QubitCircuit(2, num_cbits=1)
qc.add_gate("X", 1)
qc.add_gate("SNOT", 1)
qc.add_gate("SNOT", 0)
qc.add_gate("CNOT", 1, 0)
qc.add_gate("X", 1)
qc.add_gate("SNOT", 0)
qc.add_measurement("M0", targets=[0], classical_store=0)
```

Quantum assembler

```
OPENQASM 2.0;  
include "qelib1.inc";
```

```
qreg q[2];  
creg c[1];
```

```
x q[1];  
h q[1];  
h q[0];  
cx q[0], q[1];  
x q[1];  
h q[0];  
measure q[0] -> c[0]
```

Physical system

Schrödinger equation

$$i\hbar \frac{\partial \Psi(t)}{\partial t} = \hat{H} \Psi(t)$$

- ▶ H is a Hermitian operator (*i.e.* $H = H^\dagger$) called **Hamiltonian** of the system.
- ▶ $\Psi(t)$ is a *wave function* of the system
- ▶ $\hbar$ is the reduced Planck constant

Physical system

W

Solution to the Schrödinger equation:

$$|\psi_{t_1}\rangle = U(t_0, t_1) |\psi_{t_0}\rangle ,$$

where

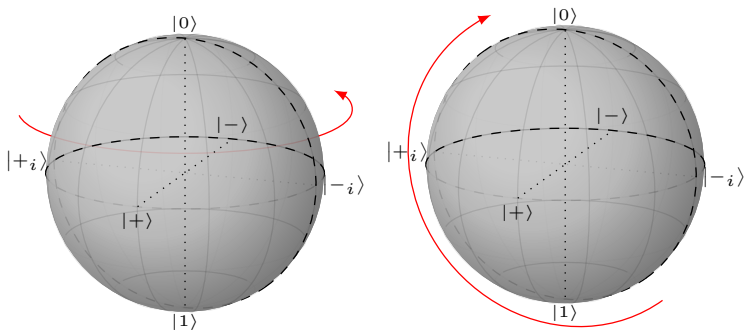
- ▶ $|\psi_{t_0}\rangle$ — initial state of the system,
- ▶ $|\psi_{t_1}\rangle$ — final state of the system
- ▶ $U(t_0, t_1)$ — **unitary operator** driving the system from t_0 to t_1 .
- ▶ An operator is unitary iff $UU^\dagger = U^\dagger U = \mathbb{1}$
- ▶ If Hamiltonian H is constant between time t_0 and t_1 , then:

$$U(t_0, t_1) = e^{-i(t_1-t_0)H},$$

where $e^M = \sum_{k=0}^{\infty} \frac{M^k}{k!}$.

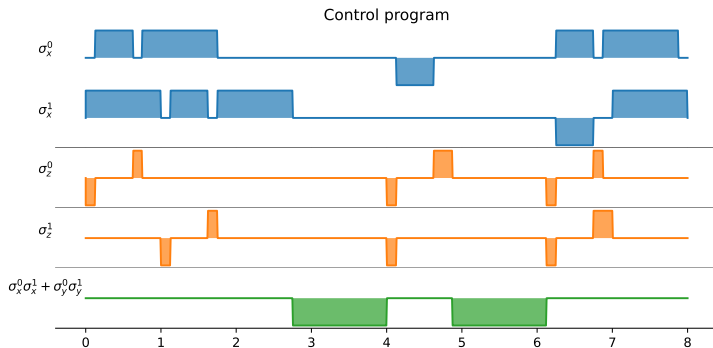
Physical system

$$H(t) = h_{x_0}(t)\sigma_x^0 + h_{x_1}(t)\sigma_x^1 + h_{z_0}(t)\sigma_z^0 + h_{z_1}(t)\sigma_z^1 + J(t) (\sigma_x^0\sigma_x^1 + \sigma_y^0\sigma_y^1)$$



Control program

$$H(t) = h_{x_0}(t)\sigma_x^0 + h_{x_1}(t)\sigma_x^1 + h_{z_0}(t)\sigma_z^0 + h_{z_1}(t)\sigma_z^1 + J(t) (\sigma_x^0\sigma_x^1 + \sigma_y^0\sigma_y^1)$$



Outline for section 8

Problem of noisy devices

- A zoo of quantum noisy channels

- Error correction

Qubit channels — zoo

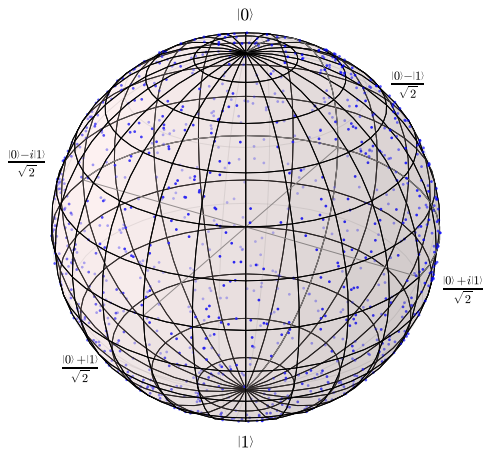


Figure: Uniformly chosen 1000 random **pure states**.

Qubit channels — zoo

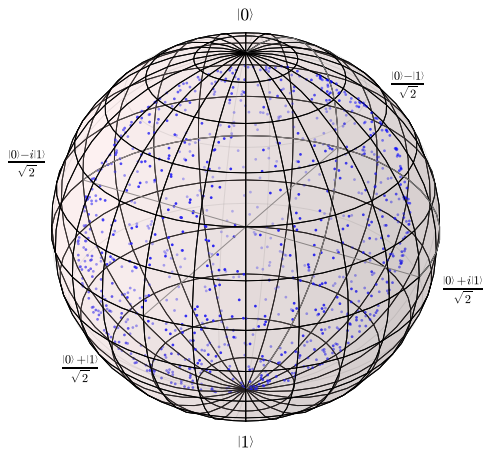


Figure: Action of **bit flip channel**: $p = 0.1$.

Qubit channels — zoo

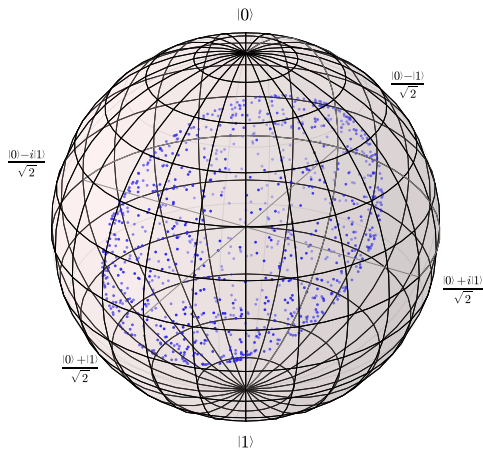


Figure: Action of bit flip channel: $p = 0.2$.

Qubit channels — zoo

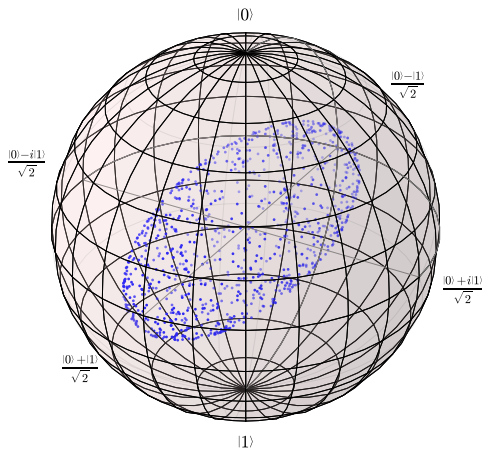


Figure: Action of bit flip channel: $p = 0.3$.

Qubit channels — zoo

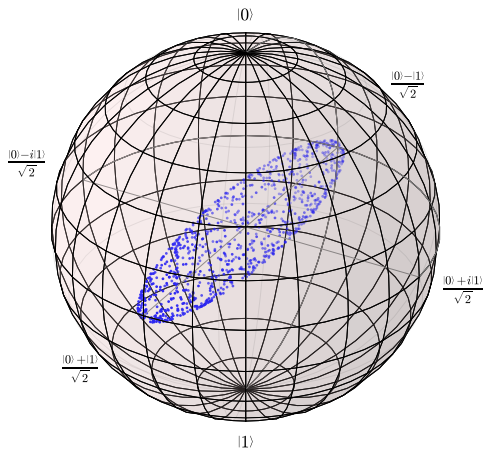


Figure: Action of bit flip channel: $p = 0.4$.

Qubit channels — zoo

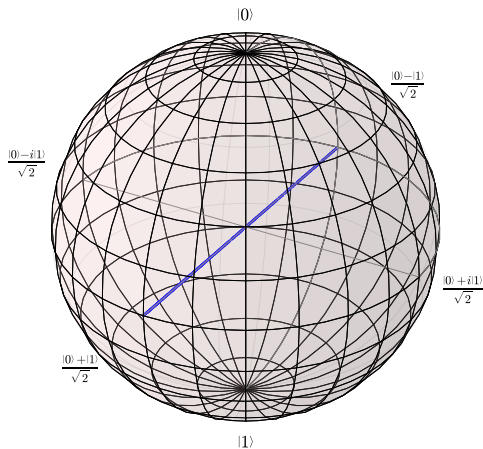


Figure: Action of bit flip channel: $p = 0.5$.

Qubit channels — zoo

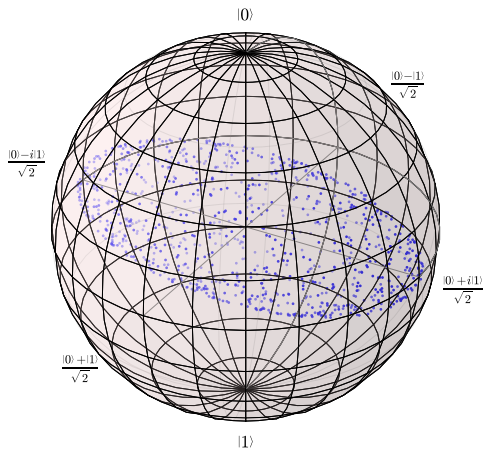


Figure: Action of **phase flip channel**: $p = 0.3$.

Qubit channels — zoo

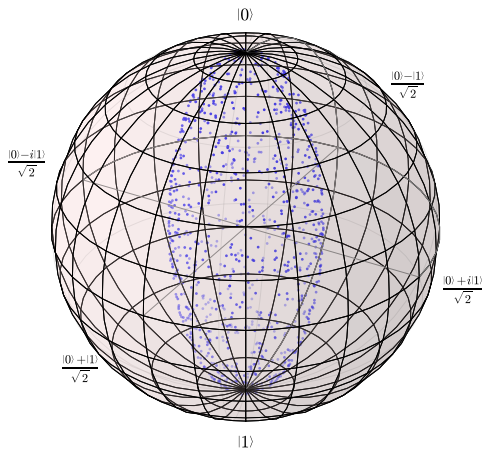


Figure: Action of **bit-phase flip channel**: $p = 0.3$.

Qubit channels — zoo

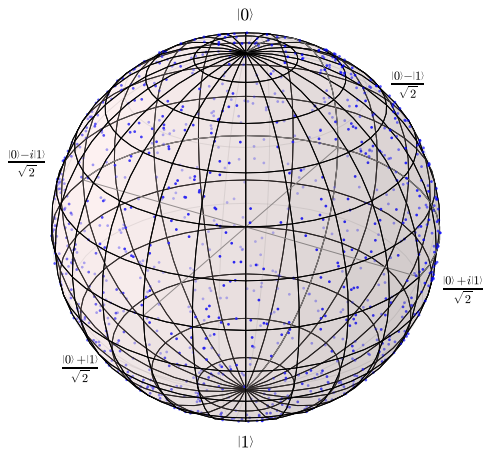


Figure: Uniformly chosen 1000 random pure states.

Qubit channels — zoo

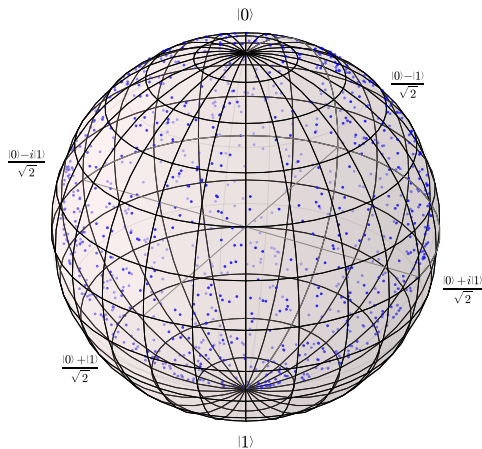


Figure: Action of amplitude **damping channel**: $p = 0.1$.

Qubit channels — zoo

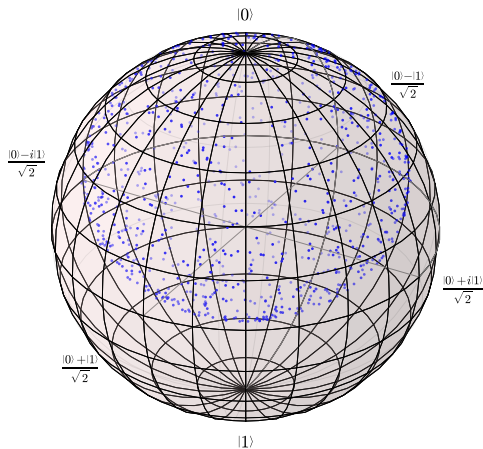


Figure: Action of amplitude damping channel: $p = 0.3$.

Qubit channels — zoo

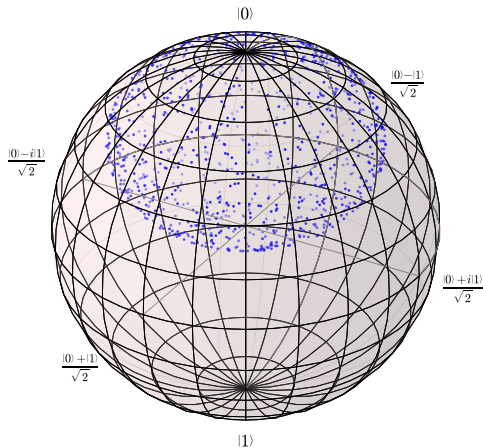


Figure: Action of amplitude damping channel: $p = 0.5$.

Qubit channels — zoo

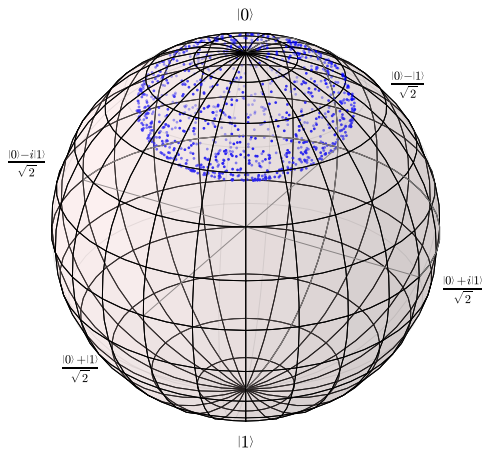


Figure: Action of amplitude damping channel: $p = 0.7$.

Qubit channels — zoo

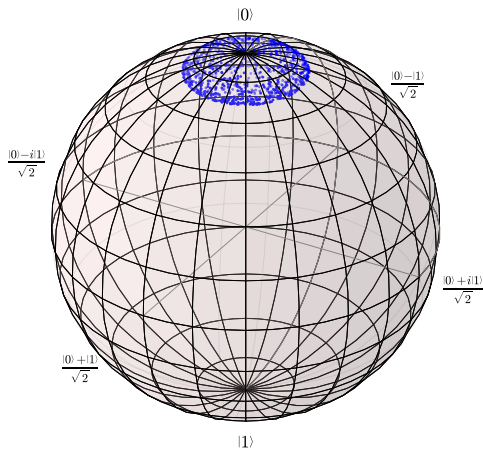
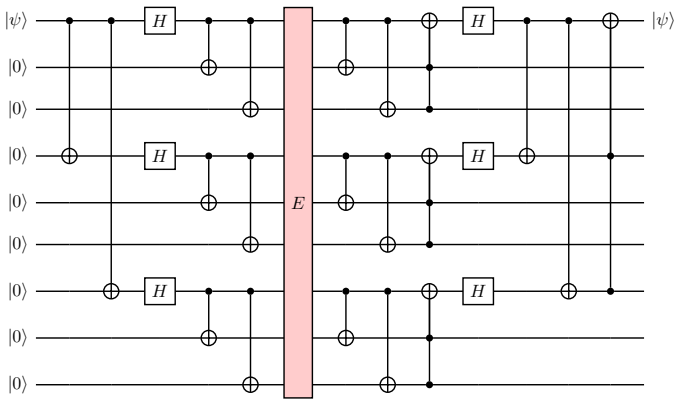


Figure: Action of amplitude damping channel: $p = 0.9$.

Shor error correction scheme



Outline for section 9

Quantum neural networks

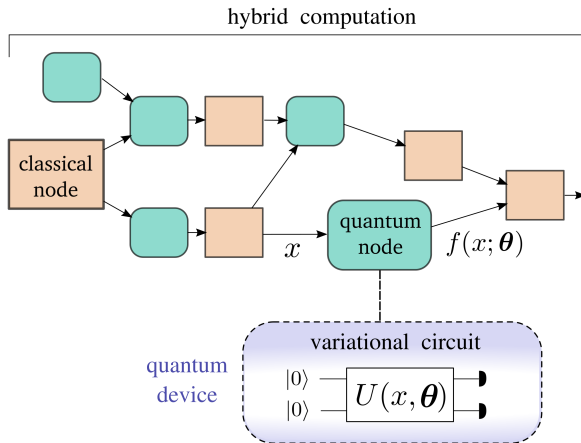
- Hybrid quantum–classical computation

- Variational quantum circuits

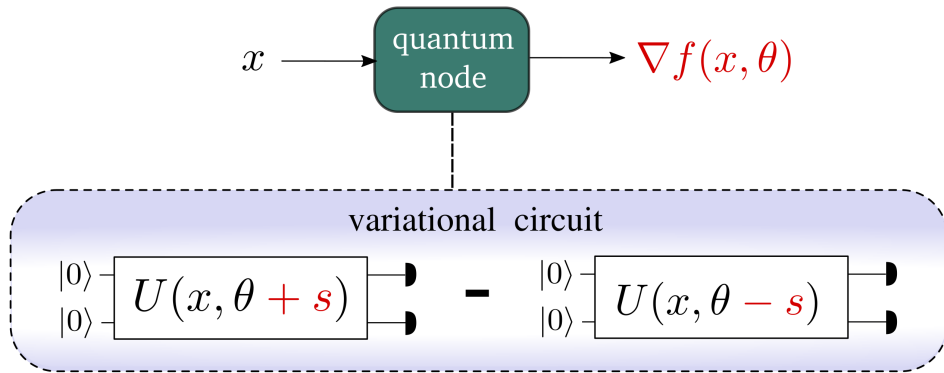
- Quantum neural network

- Mathematical tools of QNNs

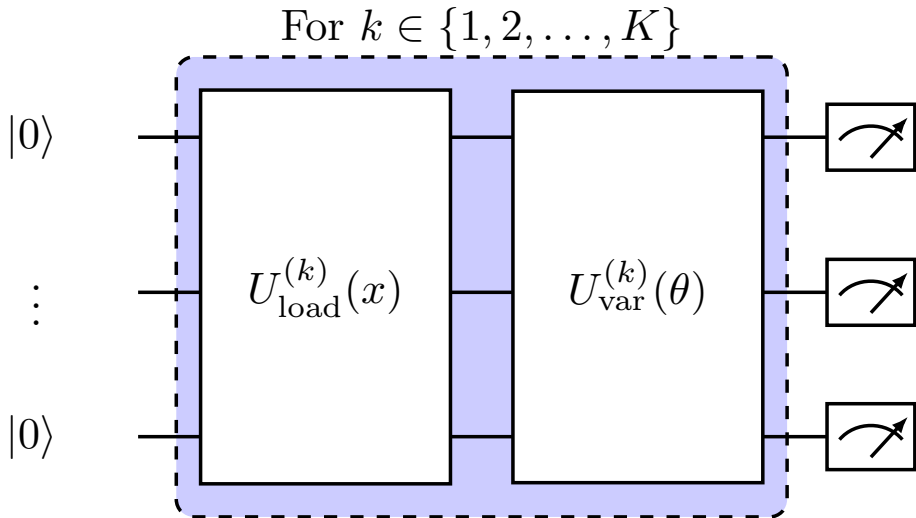
Concept of hybrid computation



Gradient — parameter shift rule



Variational quantum circuit



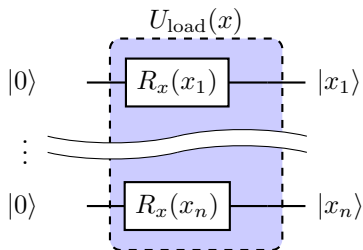
Classification with Quantum Neural Network

Data encoding

A data sample

$$x = (x_1, x_2, \dots, x_N)$$

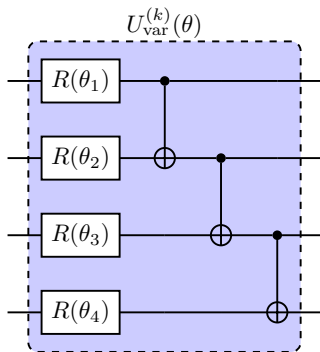
is encoded into a quantum state $|x\rangle$.



Classification with Quantum Neural Network

Variational gate

A variational gate $U_{\text{var}}(\theta)$ represents transformation of data by our quantum neural network.



Observable, expectation value

- ▶ Let O be an **observable**—a quantum measurement whose labels are real numbers.
- ▶ Mathematically it is represented as a Hermitian operator i.e.

$$O^\dagger = O.$$

- ▶ The **expectation value** of an observable O in a state $|\psi\rangle$

$$\langle O \rangle_{|\psi\rangle} = \langle \psi | O | \psi \rangle .$$

- ▶ For example if we observe spin of a system, expectation value tells us about the average spin of this system in a given state.

Gradients, parameter shift rule

- A function

$$f(\theta) = \langle O \rangle_{U(\theta)|\psi\rangle}$$

if $U(\theta) = e^{-i\theta G}$, where G has two distinctive eigenvalues $\{-r, r\}$ has gradient

$$\frac{\partial}{\partial \theta} f = r [f(\theta + s) - f(\theta - s)],$$

where $s = \pi/4r$.

- For example for $R_x(\theta) = e^{-\theta i\sigma_x/2}$, and $f(\theta) = \langle O \rangle_{R_x(\theta)|\psi\rangle}$ the gradient is

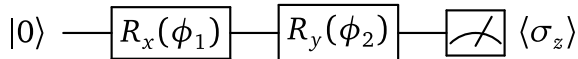
$$\frac{\partial}{\partial \theta} f(\theta) = \frac{1}{2} [f(\theta + \pi/2) - f(\theta - \pi/2)].$$

Optimizers

- ▶ Gradient descent: $\theta^{(t+1)} = \theta^{(t)} - \eta \nabla f(\theta^{(t)});$
- ▶ Nesterov momentum: $\theta^{(t+1)} = \theta^{(t)} - a^{(t+1)},$
 $a^{(t+1)} = ma^{(t)} + \eta \nabla f(\theta^{(t)} - ma^{(t)});$

With η — the step size, m — the momentum.

Variational quantum algorithm I



```
import pennylane as qml
from pennylane.optimize import GradientDescentOptimizer
from pennylane import numpy as np
# Create device
dev = qml.device('default.qubit', wires=1)
# Quantum node
@qml.qnode(dev)
def circuit1(var):
    qml.RX(var[0], wires=0)
    qml.RY(var[1], wires=0)
    return qml.expval(qml.PauliZ(0))
# Create optimizer
opt = GradientDescentOptimizer(0.25)
# Optimize circuit output
var = np.array([0.5, 0.2], requires_grad=True)
for it in range(30):
    var = opt.step(circuit1, var)
    print("Step {}: cost: {}".format(it, circuit1(var)))
```

Variational quantum algorithm II

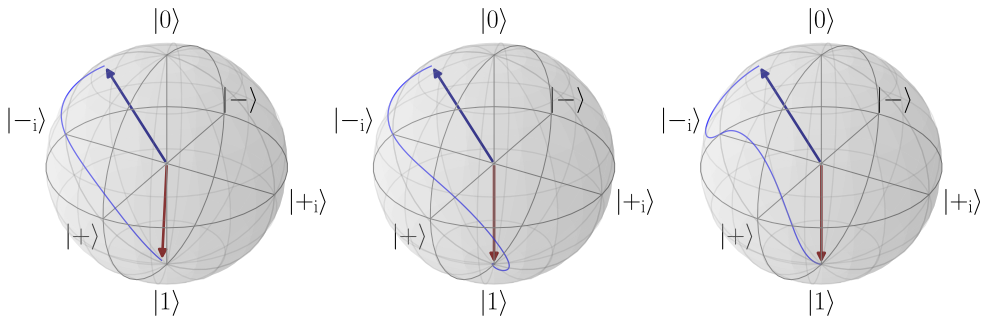


Figure: Three optimization strategies: gradient descent, Nesterov momentum optimizer, Adam optimizer ($var_{init} = (0.5, 0.2)$).

An example of quantum neural network

Dataset

We use an artificial dataset (moons) consisting of two classes.

Data preprocessing

- ▶ The original feature vectors (single feature for all samples) X_{orig} are normalized

$$X_{\text{std}} = \frac{X_{\text{orig}} - \min(X_{\text{orig}})}{\max(X_{\text{orig}}) - \min(X_{\text{orig}})}$$

$$X_{\text{scaled}} = X_{\text{std}}(\max - \min) + \min,$$

with $\min = 0$, $\max = \pi$.

- ▶ In order to encode as in angle of a qubit rotation.

An example of quantum neural network

Data encoding gate

- ▶ Encoding gate $U_e(x)$ transforms data sample $x = (x_1, x_2, \dots, x_N) \in \mathbb{R}^N$ into data state

$$|x\rangle = U_e(x) |0\rangle^{\otimes N} = R_x(x_1) \otimes R_x(x_2) \otimes \dots \otimes R_x(x_N) |0\rangle^{\otimes N},$$

- ▶ where

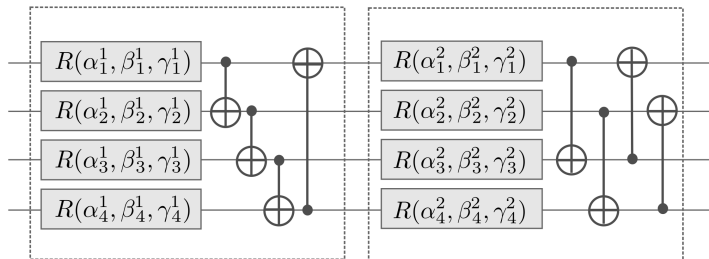
$$R_x(\phi) = e^{-i\phi\sigma_x/2} = \begin{bmatrix} \cos(\phi/2) & -i\sin(\phi/2) \\ -i\sin(\phi/2) & \cos(\phi/2) \end{bmatrix}$$

- ▶ and $x_i \in [0, \pi]$.

An example of quantum neural network

Variational gate

- Variational gate $U_v(w)$ entangles the information and is parametrized. Angles $w \in \mathbb{R}^{N \times 3 \times L}$ are the parameters we want to learn from the data and L denotes the number of quantum layers.



An example of quantum neural network

Decision function

- ▶ The decision function $f : \mathbb{R}^N \rightarrow \mathbb{R}$.
- ▶ $f(x; \theta) = \langle O \rangle_{U_v(\theta)U_e(x)|0\rangle}$, with

$$\langle O \rangle_{U_v(\theta)U_e(x)|0\rangle} = \langle 0 | U_e(x)^\dagger U_v(\theta)^\dagger O U_v(w) U_e(x) | 0 \rangle ,$$

- ▶ Observable $O = \sigma_z \otimes \mathbb{1}^{\otimes(N-1)}$.
- ▶ The classification function
 $\text{cls} : \mathbb{R}^N \rightarrow \{-1, 1\}$: $\text{cls}(x; \theta) = \text{sgn}(f(x; \theta))$.

An example of quantum neural network

Training

- ▶ The available data are divided into: train and test.
- ▶ Initial θ -s: $w \sim \mathcal{U}(0, 2\pi)$.
- ▶ Cost function

$$\text{cost} \left((y_i^{\text{train}})_{i \in \xi_t}, (f(X_{i,:}^{\text{train}}; \theta))_{i \in \xi_t} \right) = \sum_{i \in \xi_t} (y_i^{\text{train}} - f(X_{i,:}^{\text{train}}; \theta))^2 / |\xi_t|.$$

- ▶ Batches ξ_t .

Outline for section 10

Example in python

- Source code

- Visualization of learning process

Binary classification in pennylane

Imports

```
from importlib.metadata import requires
from itertools import chain

from sklearn import datasets
from sklearn.utils import shuffle
from sklearn.preprocessing import minmax_scale
from sklearn.model_selection import train_test_split
import sklearn.metrics as metrics

import pennylane as qml
from pennylane import numpy as np
from pennylane.templates.embeddings import AngleEmbedding
from pennylane.templates.layers import StronglyEntanglingLayers
```

Binary classification in pennylane

Data import, preprocessing and splitting

```
np.random.seed(42)

# create the dataset
X, y = datasets.make_moons(100, noise=0.1, random_state=42)

# shuffle the data
X, y = shuffle(X, y, random_state=42)

# normalize data
X = minmax_scale(X, feature_range=(0, np.pi))

# split data into train and test
```

Binary classification in pennylane

Building the quantum classifier

```
# number of qubits is equal to the number of features
n_qubits = X.shape[1]

# quantum device handle
dev = qml.device("default.qubit", wires=n_qubits)

# quantum circuit
@qml.qnode(dev)
def circuit(weights, x=None):
    AngleEmbedding(x, wires = range(n_qubits))
    StronglyEntanglingLayers(weights, wires = range(n_qubits))
    return qml.expval(qml.PauliZ(0))

def cost(theta, X, expectations):
    e_predicted = np.array([circuit(theta, x=x) for x in X])

    loss = np.mean((e_predicted - expectations)**2)
    return loss

# helper function: returns 1 if x>0 else 0
def sgn(x):
    return (x>0)*1
```

Binary classification in pennylane

Training preparation

```
# number of quantum layers
n_layers = 3

# convert classes to expectations: 0 to -1, 1 to +1
e_train = np.empty_like(y_train)
e_train[y_train == 0] = -1
e_train[y_train == 1] = +1

# select learning batch size
batch_size = 5

# calculate number of batches
batches = len(X_train) // batch_size

# select number of epochs
n_epochs = 5

# draw random quantum node weights
shape = qml.StronglyEntanglingLayers.shape(n_layers=n_layers, n_wires=n_qubits)
theta = np.random.random(size=shape, requires_grad=True)
```

Binary classification in pennylane

Training

```
# train the variational classifier

# start of main learning loop
# build the optimizer object
pennylane_opt = NesterovMomentumOptimizer()

log = []
# split training data into batches
X_batches = np.array_split(np.arange(len(X_train)), batches)
for it, batch_index in enumerate(chain(*(n_epochs * [X_batches]))):
    # Update the weights by one optimizer step
    batch_cost = \
        lambda t: cost(t, X_train[batch_index], e_train[batch_index])
    theta = pennylane_opt.step(batch_cost, theta)
    log.append({"theta": theta})
```

Binary classification in pennylane

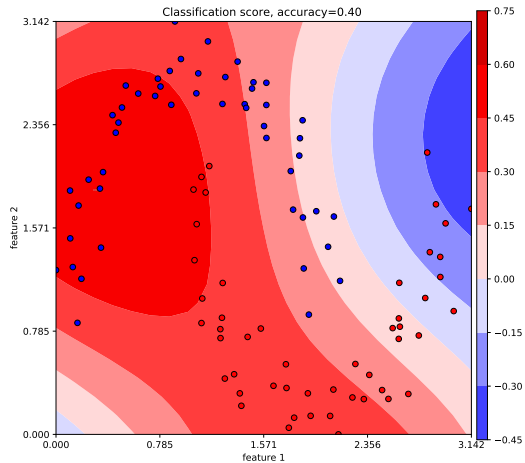
Inference

```
# convert scores to classes
scores = np.array([circuit(theta, x=x) for x in X_test])
y_pred = sgn(scores)

print(metrics.accuracy_score(y_test, y_pred))
```

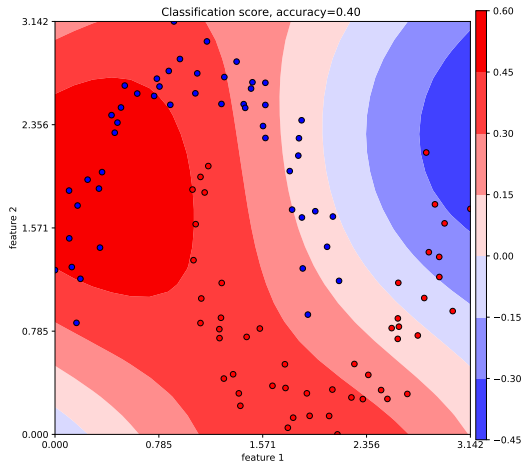
Learning process

Learning step 1



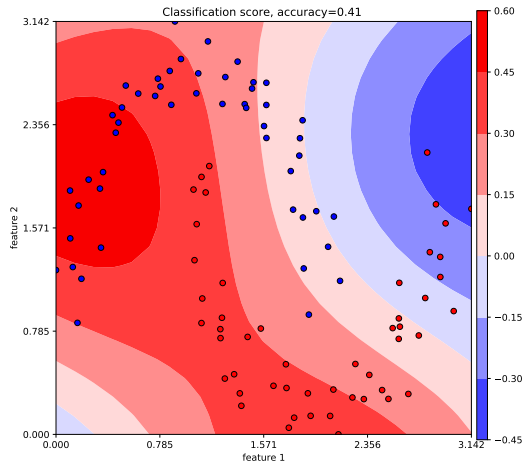
Learning process

Learning step 2



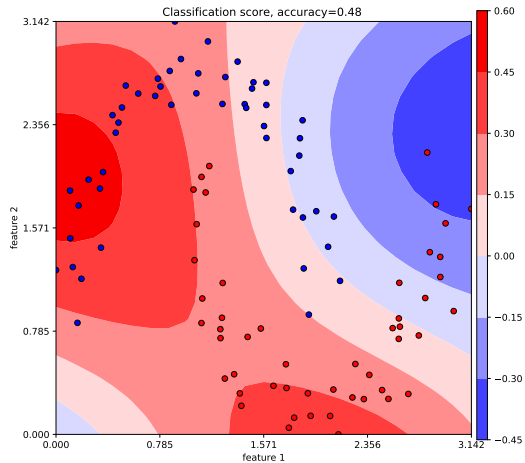
Learning process

Learning step 3



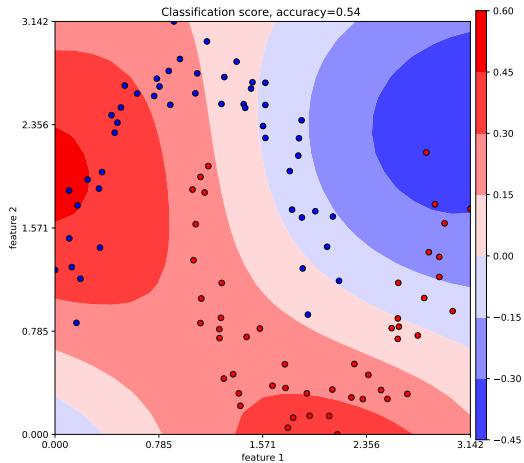
Learning process

Learning step 4



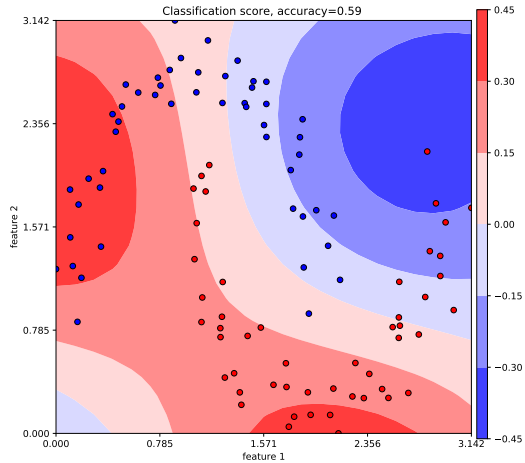
Learning process

Learning step 5



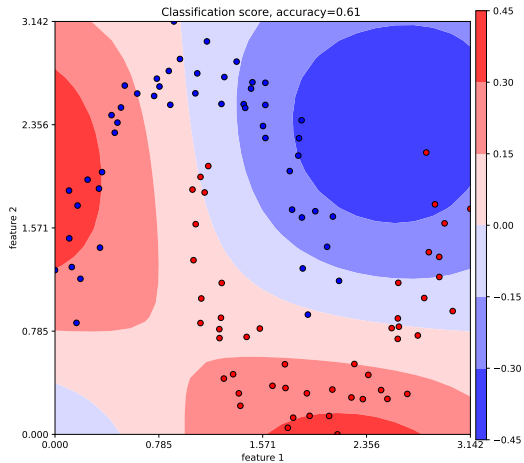
Learning process

Learning step 6



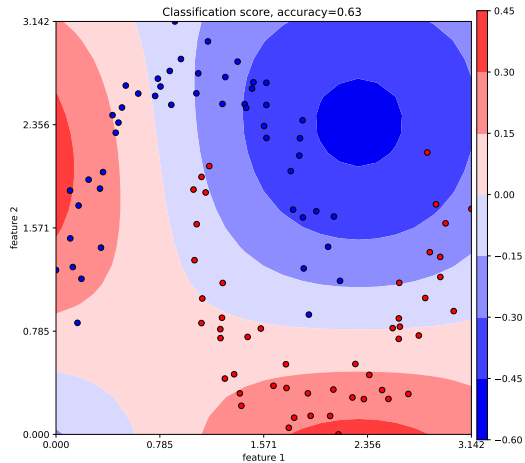
Learning process

Learning step 7



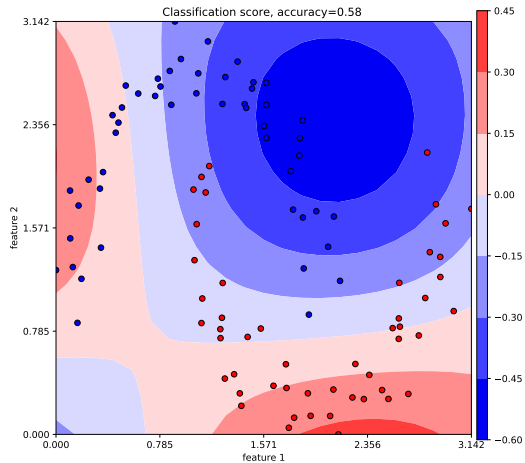
Learning process

Learning step 8



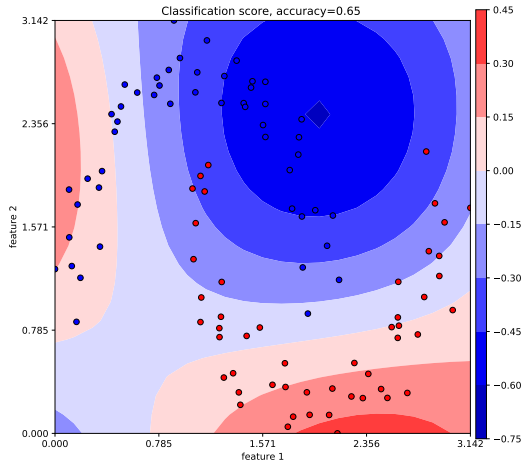
Learning process

Learning step 9



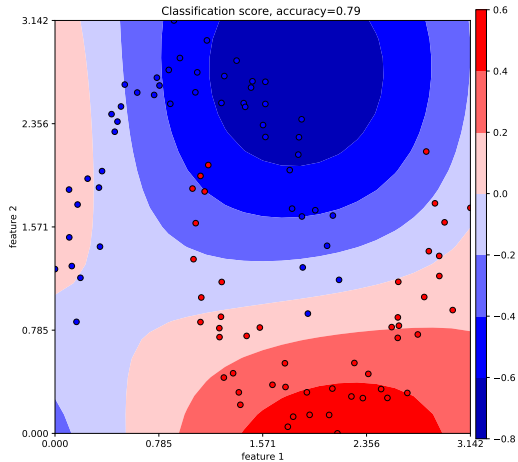
Learning process

Learning step 10



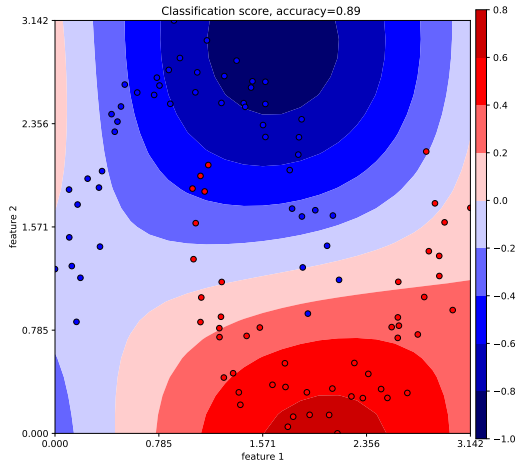
Learning process

Learning step 15



Learning process

Learning step 20



Outline for section 11

Adiabatic quantum computation

- Classical Ising model

- Ising model applications in the computer science

- Quantum Ising model

- Quantum adiabatic protocol

Classical Ising model

Let the energy function be given (Hamiltonian):

$$H(s) = - \sum_{i \in \mathcal{I}} h_i s_i - \sum_{(i,j) \in \mathcal{I} \times \mathcal{I}} J_{ij} s_i s_j,$$

where

$$s = [s_i]_{i \in \mathcal{I}} \in \{-1, 1\}^{\mathcal{I}}, h_i \in \mathbb{R}, J_{ij} \in \mathbb{R}.$$

Our goal is to find

$$s^* = \min_s H(s).$$

That is the so-called state minimum energy state.

Ising models and NP class problems

1 Introduction

- 1.1 Quantum Adiabatic Optimization
- 1.2 Ising Spin Glasses
- 1.3 The Goal of This Paper
- 1.4 What Problems Are Easy (to Embed) on Experimental AQO Devices?

2 Partitioning Problems

- 2.1 Number Partitioning
- 2.2 Graph Partitioning
- 2.3 Cliques
- 2.4 Reducing N to $\log N$ Spins in Some Constraints

3 Binary Integer Linear Programming

4 Covering and Packing Problems

- 4.1 Exact Cover
- 4.2 Set Packing
- 4.3 Vertex Cover
- 4.4 Satisfiability
- 4.5 Minimal Maximal Matching

5 Problems with Inequalities

- 5.1 Set Cover
- 5.2 Knapsack with Integer Weights

Ising models and NP class problems

6 Coloring Problems

- 6.1 Graph Coloring
- 6.2 Clique Cover
- 6.3 Job Sequencing with Integer Lengths

7 Hamiltonian Cycles

- 7.1 Hamiltonian Cycles and Paths
- 7.2 Traveling Salesman

8 Tree Problems

- 8.1 Minimal Spanning Tree with a Maximal Degree Constraint
- 8.2 Steiner Trees
- 8.3 Directed Feedback Vertex Set
- 8.4 Undirected Feedback Vertex Set
- 8.5 Feedback Edge Set

9 Graph Isomorphisms

10 Conclusions

References

Quantum (simplified) Ising model

Problem Hamiltonian

We have a given problem Hamiltonian

$$H_p = - \sum_{i \in \mathcal{I}} h_i \sigma_z^{(i)} - \sum_{(i,j) \in \mathcal{I} \times \mathcal{I}} J_{ij} \sigma_z^{(i)} \sigma_z^{(j)},$$

where

$$\sigma_{\{x,z\}}^{(i)} = \mathbb{1}_2^{\otimes (i-1)} \otimes \sigma_{\{x,z\}} \otimes \mathbb{1}_2^{\otimes (|\mathcal{I}|-i-1)}$$
$$\mathbb{1}_2 = / \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Quantum adiabatic protocol I

Initial Hamiltonian

We have a given initial Hamiltonian

$$H_0 = - \sum_{i \in \mathcal{I}} h_i \sigma_x^{(i)}.$$

Effective variable in time Hamiltonian

$$H(t) = (1 - \frac{t}{\tau})H_0 + \frac{t}{\tau}H_p$$

Quantum adiabatic protocol II

Hamiltonian eigenstates

Each Hamiltonian has its energies E_n ($E_0 \leq E_1 \leq \dots \leq E_n$) and respective eigenstates $|\psi\rangle_n$:

$$H|\psi\rangle_n = E_n|\psi\rangle_n.$$

If we start in the state

$$|\psi^{(0)}\rangle_0 : H_0|\psi^{(0)}\rangle_0 = E_0^{(0)}|\psi^{(0)}\rangle_0 = \left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right)^{\otimes |\mathcal{I}|},$$

then for large τ we will end in a state:

$$|\psi^{(p)}\rangle_0 : H_p|\psi^{(p)}\rangle_0 = E_0^{(p)}|\psi^{(p)}\rangle_0.$$

Quantum adiabatic protocol III

In the end, we measure

$$\left\{P_{\pm 1}^{\otimes |I|}\right\},$$

gdzie

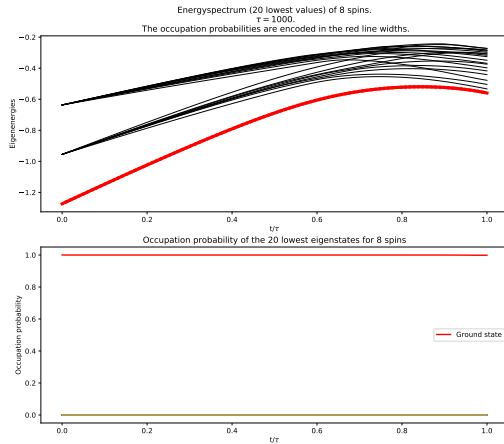
$$\{P_{-1} = |0\rangle\langle 0|, P_1 = |1\rangle\langle 1|\}.$$

As a result of the measurement, we get a string

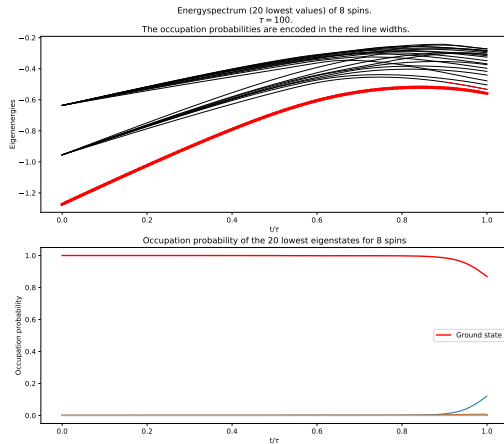
$$s = \{\pm 1\}^{|I|},$$

which is a solution to our optimization problem.

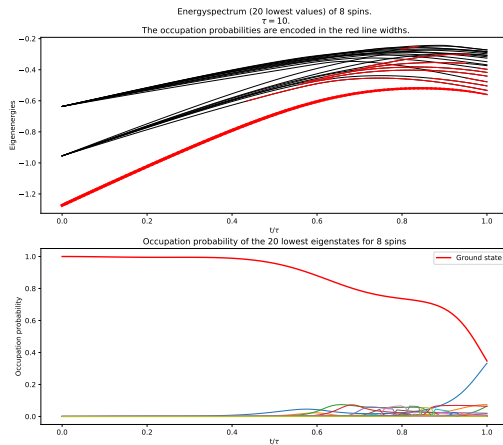
Adiabatic evolution example I



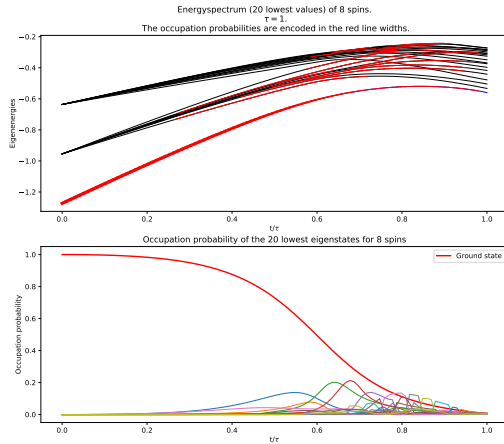
Adiabatic evolution example II



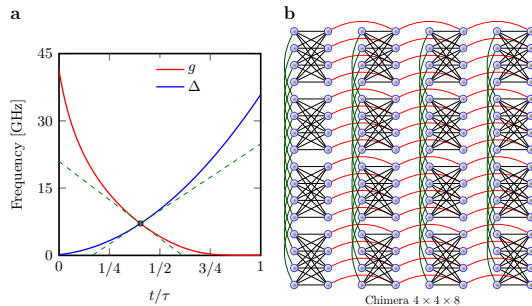
Adiabatic evolution example III



Adiabatic evolution example IV



D-Wave annealer 2000Q — Chimera architecture



$$H(t)/(2\pi\hbar) = -g(t) \sum_i \sigma_x^{(i)} - \Delta(t) H_p, \quad t \in [0, \tau].$$

Outline for section 12

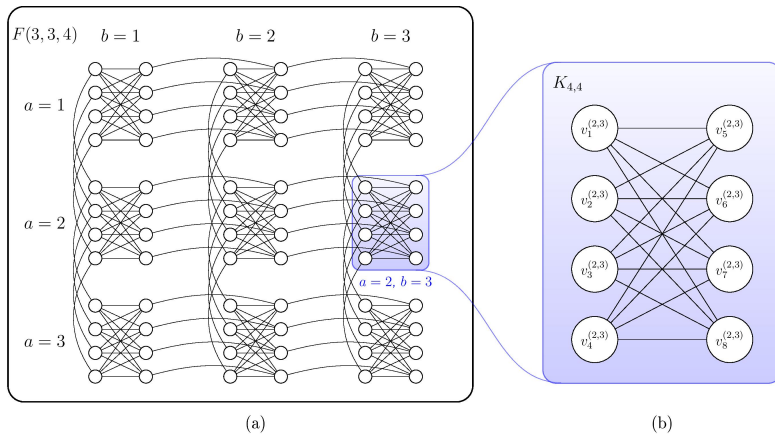
Application of D-Wave annealer for spectral image classification post-processing

- Chimera architecture

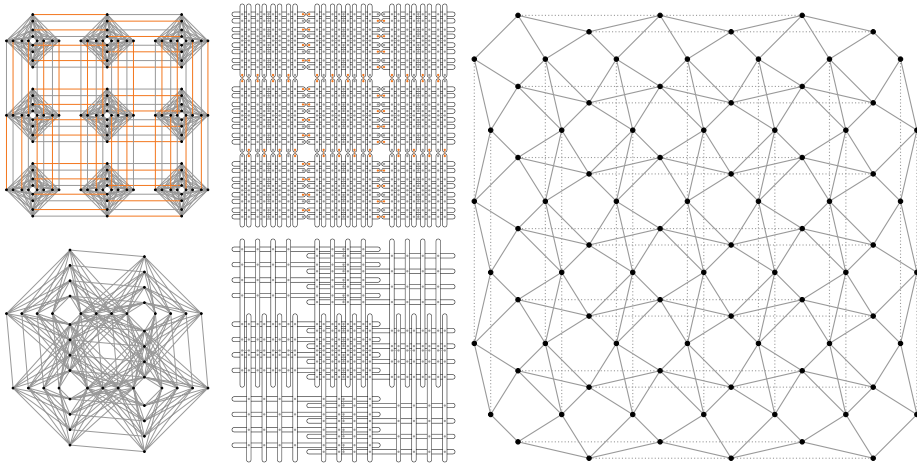
- D-Wave Zephyr architecture

- Problems with Ising solvers

Dwave-2000Q architecture



New D-Wave topology — Zephyr¹



¹Boothby2021.

QUBO/Ising optimization arms race

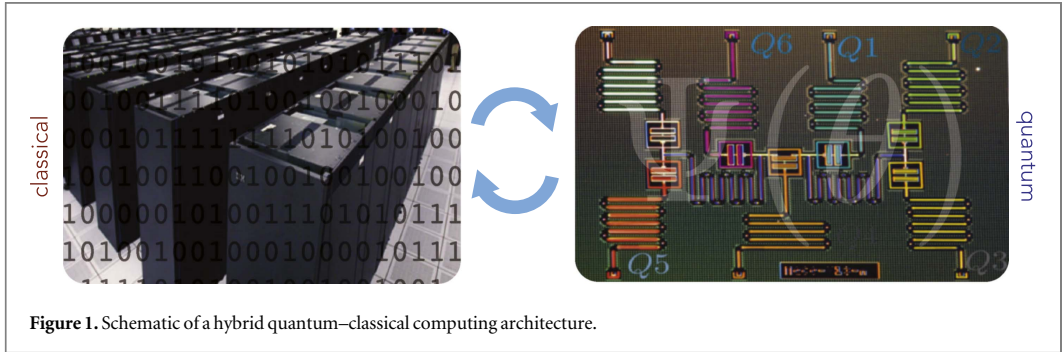
Challenges

- ▶ There is no proof that D-Wave machines are better than classical computers.
- ▶ New classical and quantum algorithms and devices are being proposed.
- ▶ Quantum Adiabatic Optimization Algorithm on gate model-based devices.
- ▶ Can we deal with noise in quantum annealers?

Applications of variational algorithms

- ▶ Finding ground states of Hamiltonians
 - ▶ Variational Quantum Eigensolver,
 - ▶ Quantum Adiabatic Optimization Algorithm.
- ▶ Machine learning — Quantum Neural Networks, fitting model to data.

The idea of variational quantum computing



Outline for section 13

Applications of Variational Quantum Eigensolvers

- Ground state energy problem

- Ground states of Fermionic Hamiltonians

- Quantum Adiabatic Optimization Algorithm

- Max-Cut problem and QAOA

Variational Quantum Eigensolver

On a NISQ quantum computer

- ▶ we can prepare $|\Psi(\theta)\rangle$ using a parametrized quantum circuit $|\Psi(\theta)\rangle = U(\theta_D)U(\theta_{K-1}) \dots U(\theta_1) |0\rangle$.
- ▶ and calculate function $E_q(\theta) = \langle \Psi(\theta) | H_q | \Psi(\theta) \rangle$.

On a classical computer

- ▶ we can vary parameters θ .

In order to

- ▶ find ground state energy $E_q^{\min} = \min_{\theta} (\langle \Psi(\theta) | H_q | \Psi(\theta) \rangle)$.

Variational Quantum Eigensolver

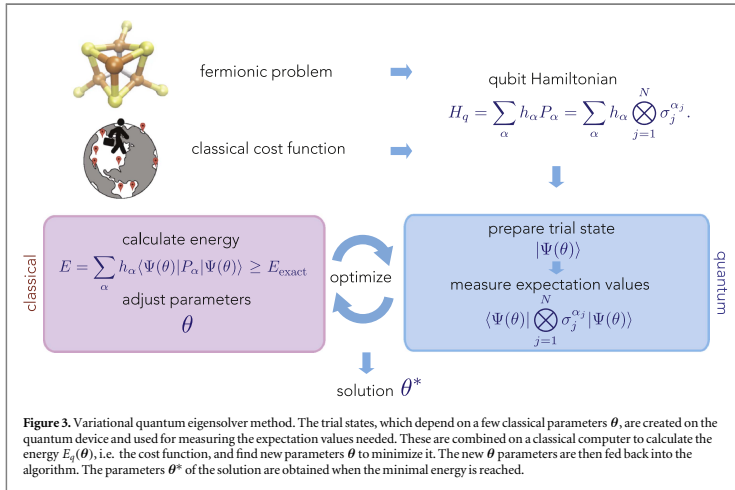
Requirements for implementation of VQ algorithm

- ▶ Small number of parameters θ .
- ▶ Able to explore the problem Hilbert space.
- ▶ Gates $U(\theta_i)$ implemented in hardware.

Problem Hamiltonian

- ▶ $H_q = \sum_{\alpha} h_{\alpha} P_{\alpha}$ — Hamiltonian can be expressed as linear combination of
- ▶ Pauli strings: $P_{\alpha} = \sigma_1^{\alpha_1} \otimes \sigma_2^{\alpha_2} \otimes \dots \otimes \sigma_N^{\alpha_N}$.
- ▶ Where $\sigma_i^j \in \{\mathbb{1}, \sigma_i^x, \sigma_i^y, \sigma_i^z\}$.
- ▶ Then energy function: $E_q(\theta) = \langle \Psi(\theta) | H_q | \Psi(\theta) \rangle = \sum_{\alpha} h_{\alpha} \langle \Psi(\theta) | P_{\alpha} | \Psi(\theta) \rangle$.

Variational quantum circuit



Another picture of variational circuit

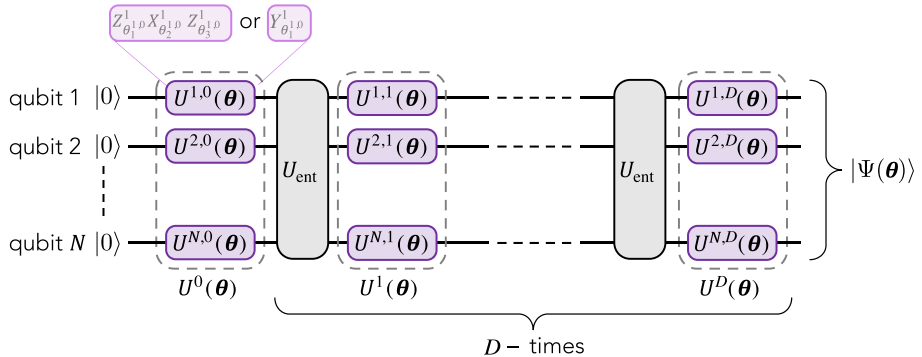


Figure 4. Heuristic preparation of trial states for the variational quantum eigensolver based on single-qubit gates $U(\theta)$ interleaved by entangling operations U_{ent} as described in the text.

Application of VQE to quantum chemistry

Electronic Hamiltonian in second quantization

- ▶ Fermionic hamiltonian $H_F = \sum_{ij} t_{ij} a_i^\dagger a_j + \sum_{ijkl} t_{ij} a_i^\dagger a_k^\dagger a_l a_j$
- ▶ with $a_i^\dagger$ creation and a_i annihilation operators.
- ▶ As an example of mapping of Fermionic Hamiltonian to qubits:
- ▶ $H_{H^2} = f_0 \mathbb{1} \otimes \mathbb{1} + f_1 \sigma_z \otimes \sigma_z + f_2 \sigma_z \otimes \mathbb{1} + f_3 \mathbb{1} \otimes \sigma_z + f_4 \sigma_z \otimes \sigma_z$
- ▶ where f_i are real coefficients dependent of the distance between atoms.

Finding ground state of Fermionic Hamiltonian

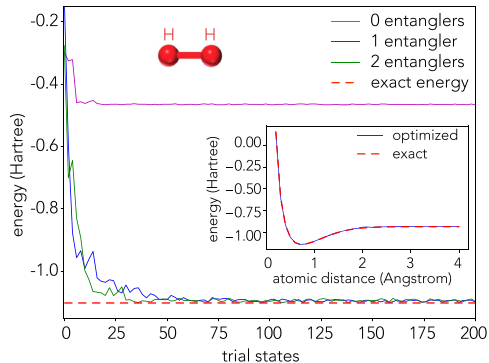


Figure 5. Quantum simulation of the hydrogen molecule on an ideal quantum simulator. At the equilibrium geometry and no entangler block in the circuit, the energy converges to a state with an energy that is about 50% too high. With two or more entanglers, the exact energy is obtained. The inset shows the entire dissociation profile for a hydrogen molecule calculated with four entangling steps.

Quantum Adiabatic Optimization Algorithm

- ▶ Cost function $C(x)$, $x_i \in \{0, 1\}$ we want to optimize.
- ▶ Encoding of C into Hamiltonian H_C .
 1. $x_i \rightarrow (1 - z_i)/2$ with $z_i \in \{-1, 1\}$,
 2. $z_i \rightarrow \sigma_i^z$
 3. $H_C = \sum_{\alpha} P_{\alpha}$
 4. where $P_{\alpha} = \sigma_1^{\alpha_1} \otimes \sigma_2^{\alpha_2} \otimes \dots \otimes \sigma_N^{\alpha_N}$ — “diagonal” Pauli strings.
 5. where $\sigma_i^j \in \{\mathbb{1}, \sigma_i^z\}$.
- ▶ Mixing Hamiltonian $H_M = -\sigma_i^x$ — transverse Hamiltonian.
- ▶ To find ground state of H_C — minimal value of $C(x)$:
- ▶ Apply $U(\beta, \gamma) = \prod_{l=1}^D e^{-i\beta_l H_M} e^{-i\gamma_l H_C}$ to a quantum state $|\psi_0\rangle$ and optimize γ, β using VQE.

Max-Cut problem

- ▶ Given a graph with the edge weights $w_{ij} : w_{ij} > 0, w_{ij} = w_{ji}$.
- ▶ Divide the graph into two disjoint subsets of vertices such that sum of weights between them is maximal.
- ▶ We want to maximize $C(x) = \sum_{ij} w_{ij} x_i (1 - x_j)$, $x_i \in \{0, 1\}$.
- ▶ $H_I = \sum_{i < j} w_{ij} (1 - \sigma_i^z)(1 + \sigma_j^z) = -\frac{1}{2} \sum_{i < j} w_{ij} \sigma_i^z \sigma_j^z + \text{const.}$
- ▶ Effectively: $H_C = \sum_{i < j} w_{ij} \sigma_i^z \sigma_j^z$.

Max-Cut using QAOA

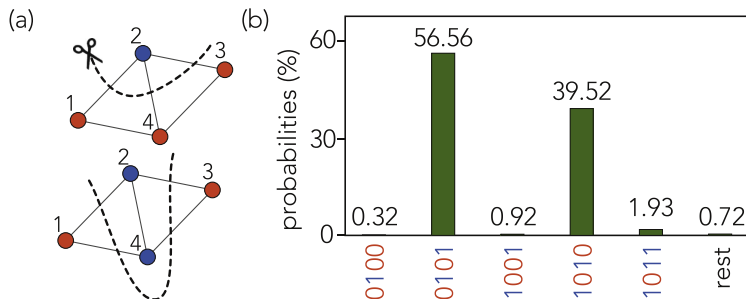


Figure 6. (a) Two different cuts for a MaxCut instance with four vertices. The lower cut has a larger cumulated weight and represents the partitioning that solves the problem. (b) Solution of the problem using the VQE with heuristic trial-states and a depth $D = 3$ circuit.